



SOICT



Multimodal Machine Learning

Lecture 4.2: Aligned Representations

Dam Quang Tuan

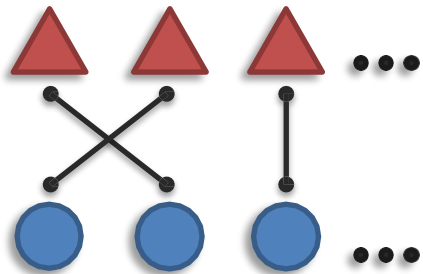
Continuous Alignment

Challenge 2: Alignment

Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

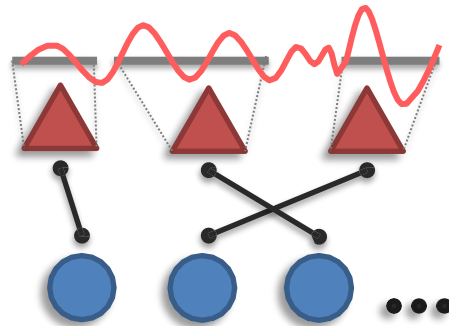
Sub-challenges:

Discrete Alignment



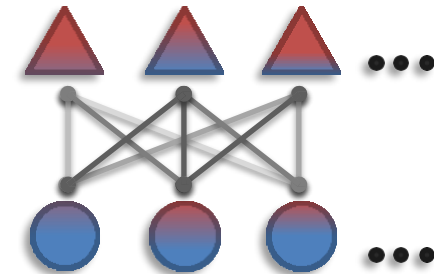
Discrete elements
and connections

Continuous Alignment



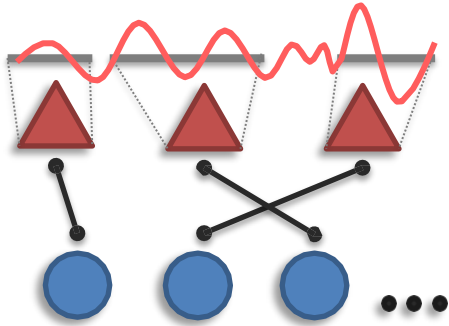
Segmentation and
continuous warping

Contextualized Representation



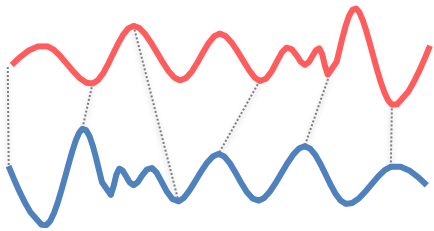
Alignment + representation

Challenge 2b: Continuous Alignment

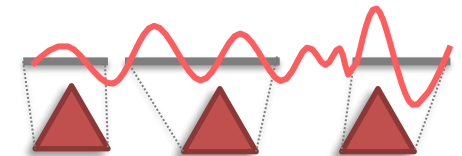


Definition: Model alignment between modalities with continuous signals and no explicit elements

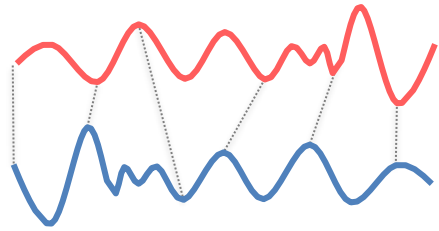
Continuous
warping



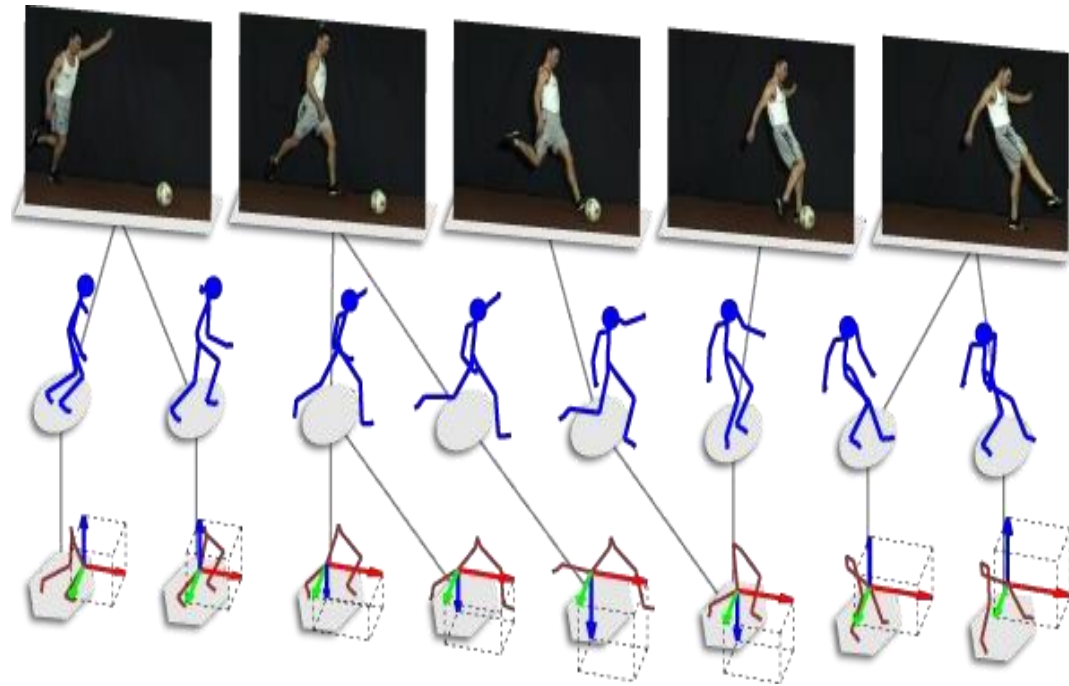
Discretization
(segmentation)



Continuous Warping – Example



➡ Aligning video sequences



Dynamic Time Warping (DTW)

We have two unaligned temporal unimodal signals

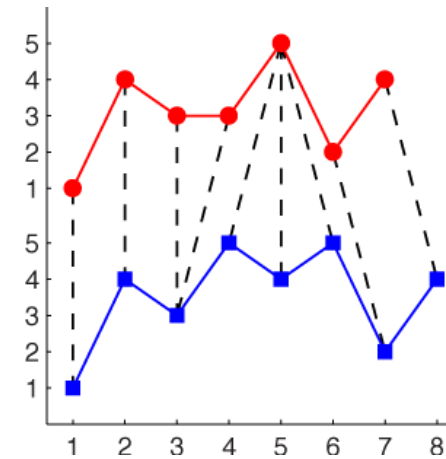
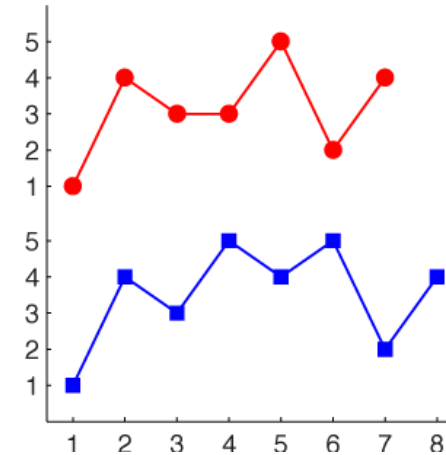
- $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n_x}] \in \mathbb{R}^{d \times n_x}$
- $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{n_y}] \in \mathbb{R}^{d \times n_y}$

Find set of indices to minimize the alignment difference:

$$L(\mathbf{p}^x, \mathbf{p}^y) = \sum_{t=1}^l \|\mathbf{x}_{p_t^x} - \mathbf{y}_{p_t^y}\|_2^2$$

where $\mathbf{p}^x$ and $\mathbf{p}^y$ are index vectors of same length

Dynamic Time Warping is designed to find these index vectors!



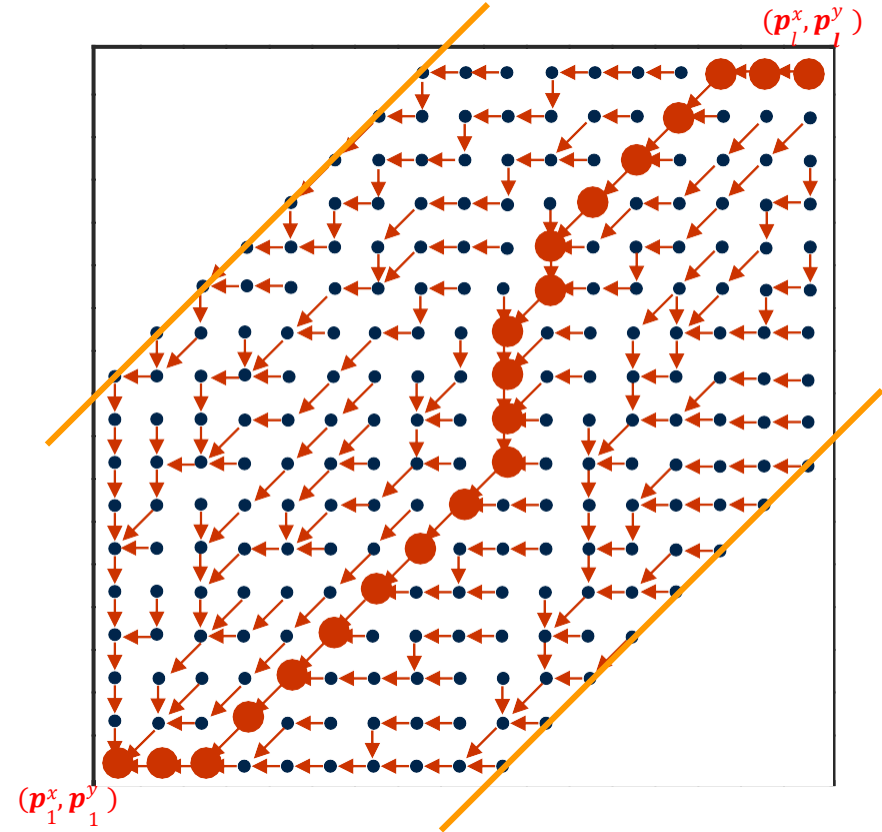
Dynamic Time Warping (DTW)

Lowest cost path in a cost matrix

- Restrictions?

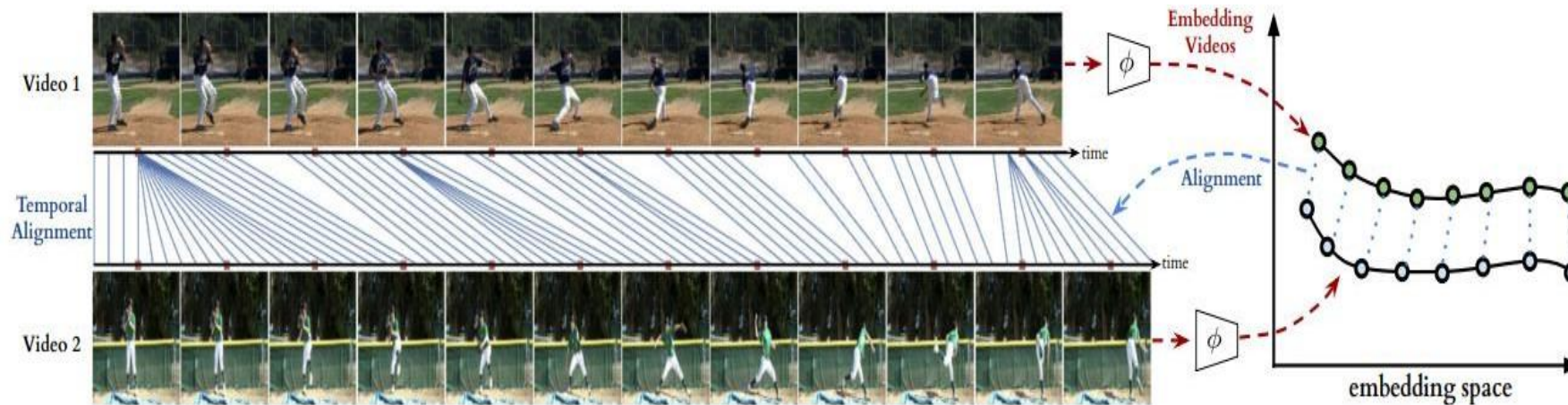
- Monotonicity – no going back in time
- Continuity - no gaps
- Boundary conditions - start and end at the same points
- Warping window - don't get too far from diagonal
- Slope constraint – do not insert or skip too much

Solved using dynamic programming while respecting the restrictions



Temporal Alignment and Neural Representation Learning

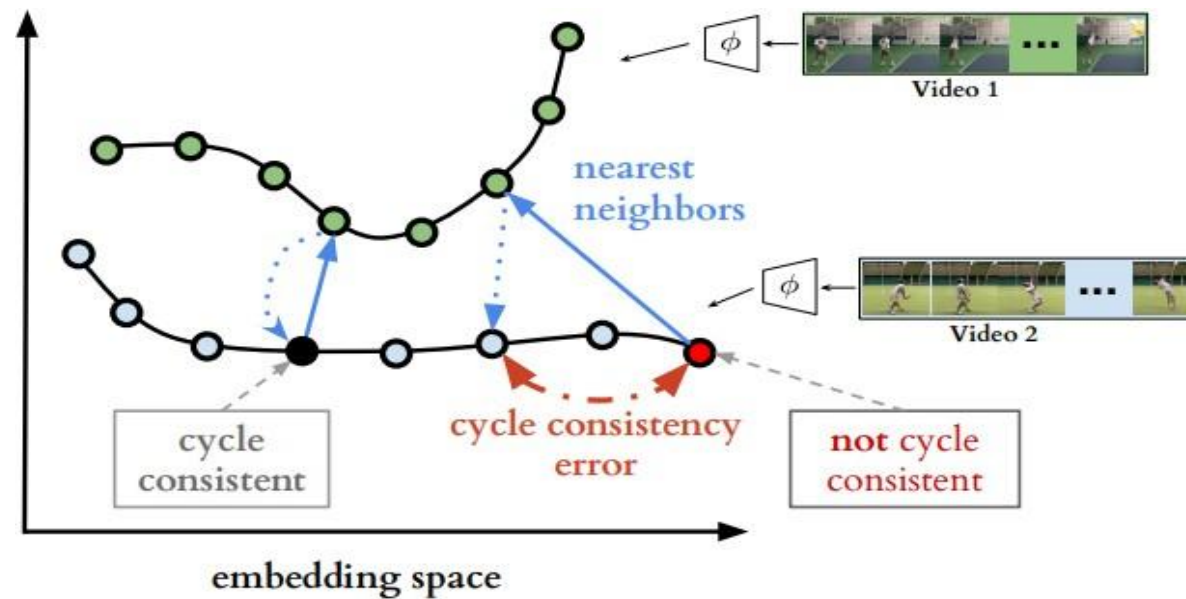
Premise: we have paired video sequences that can be temporally aligned



How can we define a loss function to enforce the alignment between sequences while at the same time learning good representations?

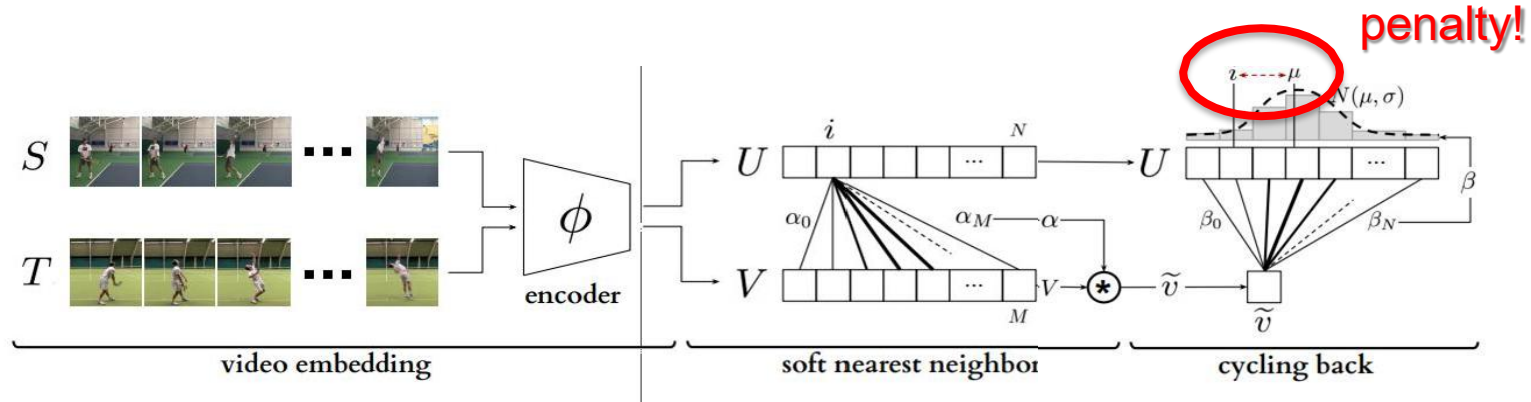
Temporal Cycle-Consistency Learning

Solution: Representation learning by enforcing **Cycle consistency**



Main idea: My closest neighbor also views me as their closest neighbor

Temporal Cycle-Consistency Learning



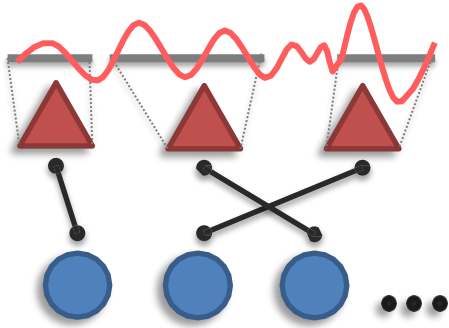
Compute “soft” / “weighted” nearest neighbour:

distances: $\alpha_j = \frac{e^{-||u_i - v_j||^2}}{\sum_k^M e^{-||u_i - v_k||^2}}$ Soft nearest neighbor: $\tilde{v} = \sum_j^M \alpha_j v_j,$

Find the nearest neighbor the other way and then penalize the distance:

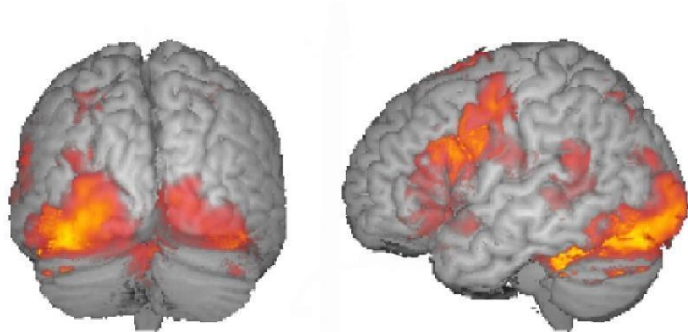
$$\beta_k = \frac{e^{-||\tilde{v} - u_k||^2}}{\sum_j^N e^{-||\tilde{v} - u_j||^2}} \quad L_{cbr} = \frac{|i - \mu|^2}{\sigma^2} + \lambda \log(\sigma)$$

Discretization (aka Segmentation)

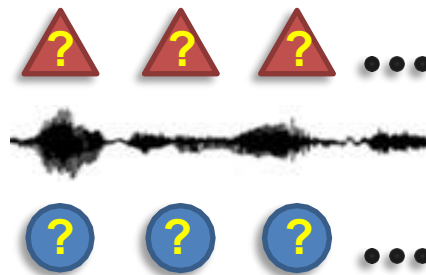


Common assumptions: ① Segmented elements

Examples:



Medical imaging



Signals

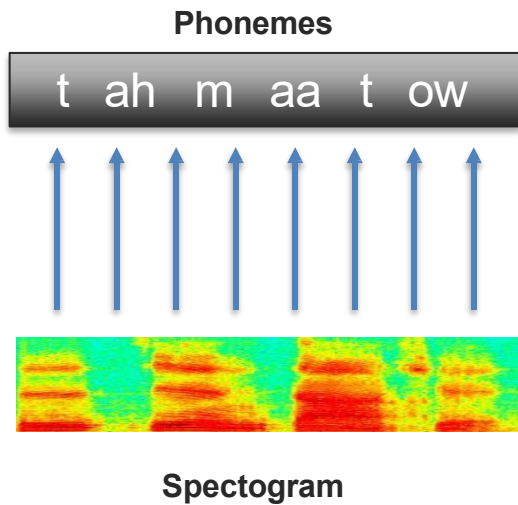


Images

objects

Discretization – Example

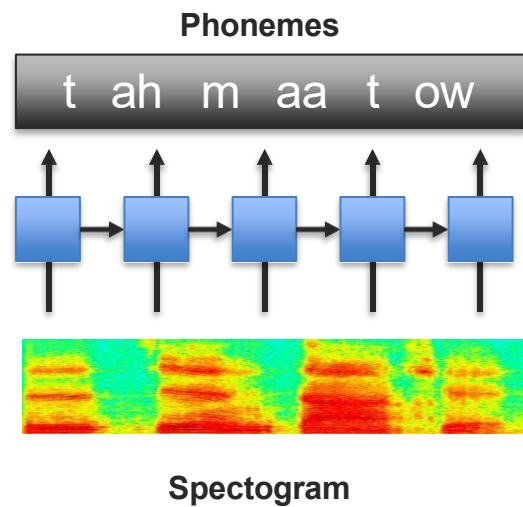
Sequence Labeling and Alignment



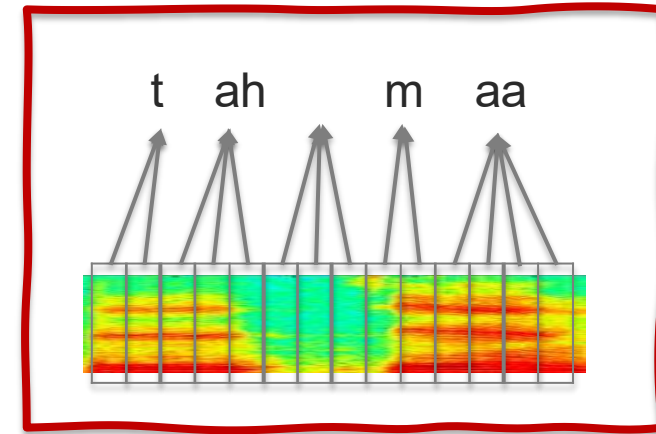
How can we predict the sequence
of phoneme labels?

Discretization – Example

Sequence Labeling and Alignment



Challenge: many-to-1 alignment

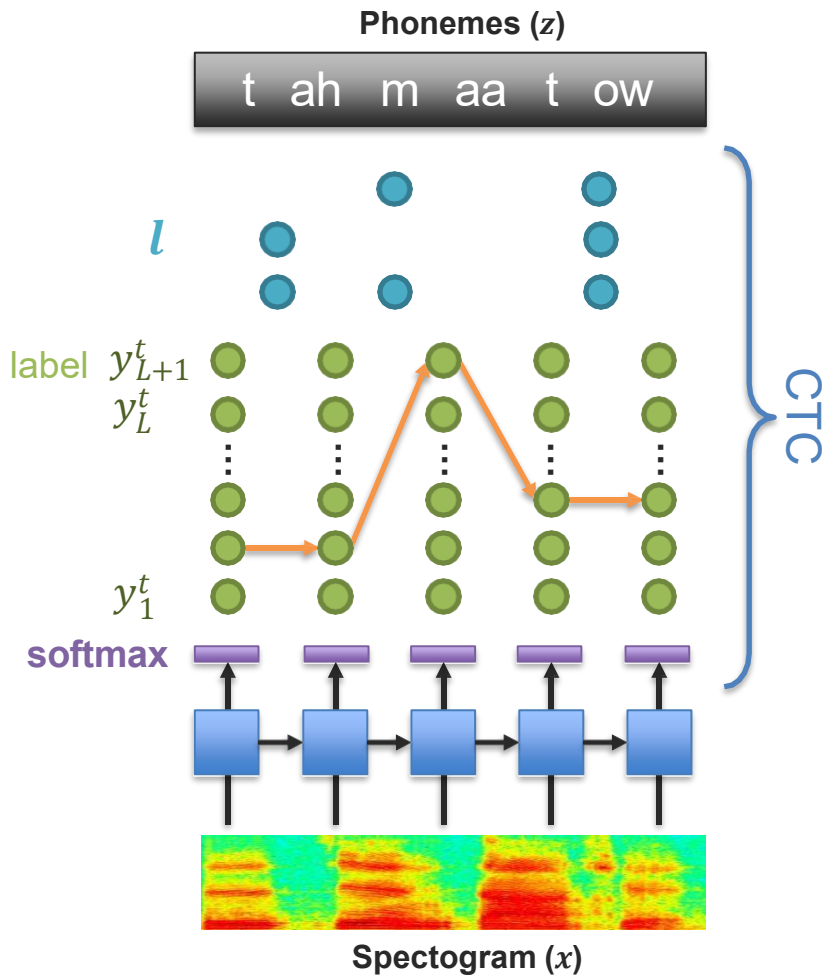


How can we predict the sequence
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Discretization – A Classification Approach

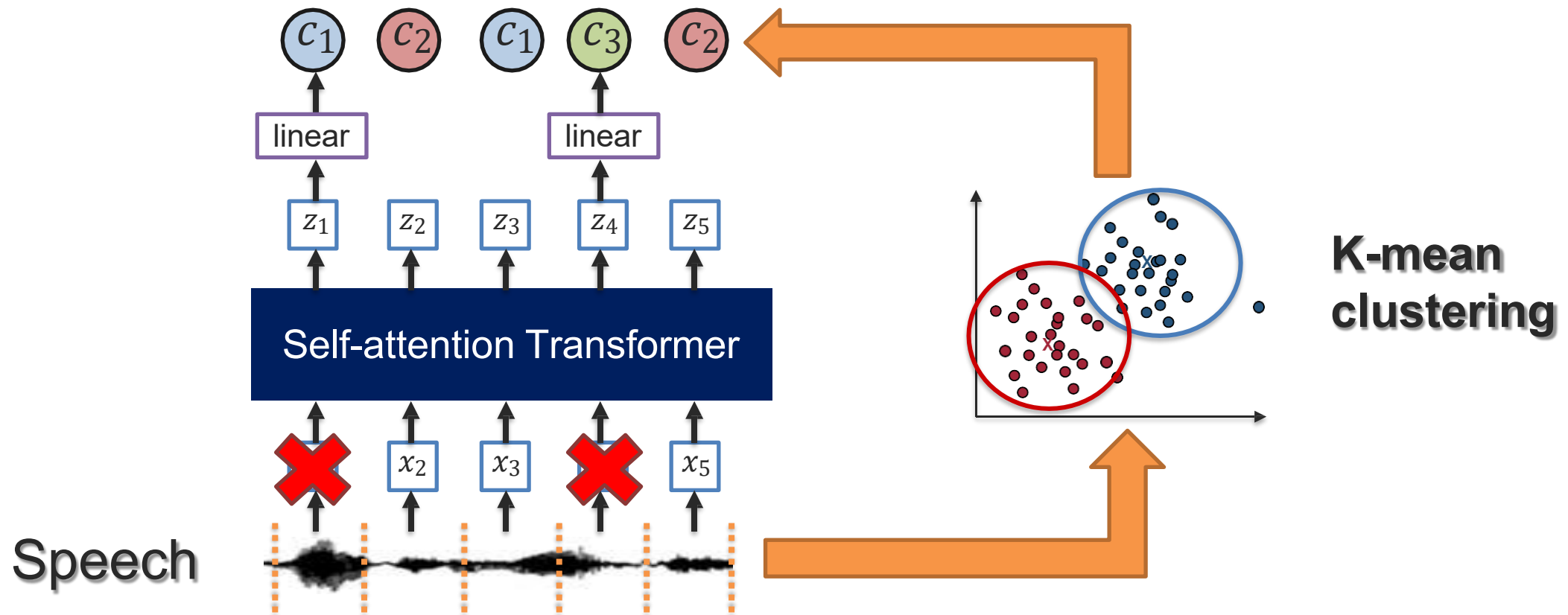
Connectionist Temporal Classification

- ④ Most probable sequence labels
- ③ Predicted labels l
- ② Path π over the activations:
- ① Output activations (distribution):



Discretization and Representation – Cluster-based Approaches

HUBERT: Hidden-Unit BERT

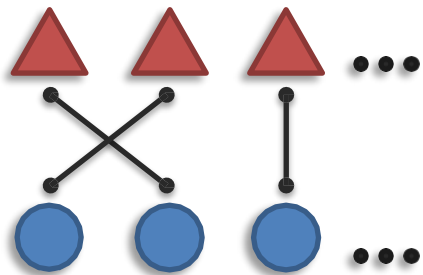


Challenge 2: Alignment

Definition: Identifying and modeling cross-modal connections between all elements of multiple modalities, building from the data structure

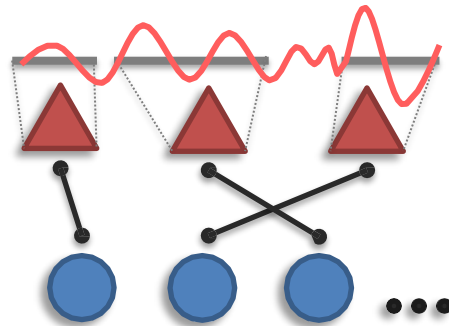
Sub-challenges:

Discrete Alignment



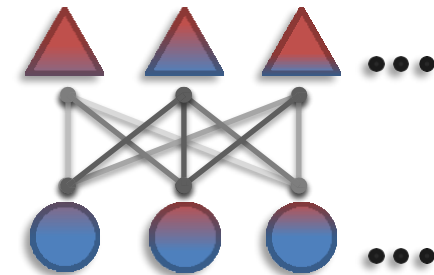
Discrete elements
and connections

Continuous Alignment



Segmentation and
continuous warping

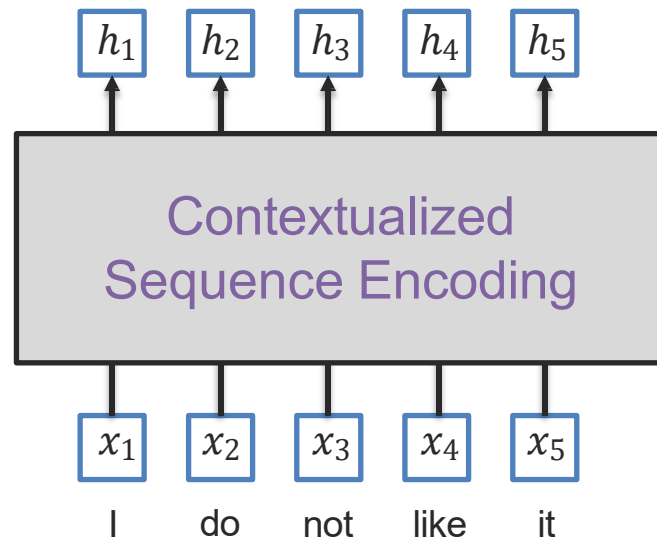
Contextualized Representation



Alignment
+ representation

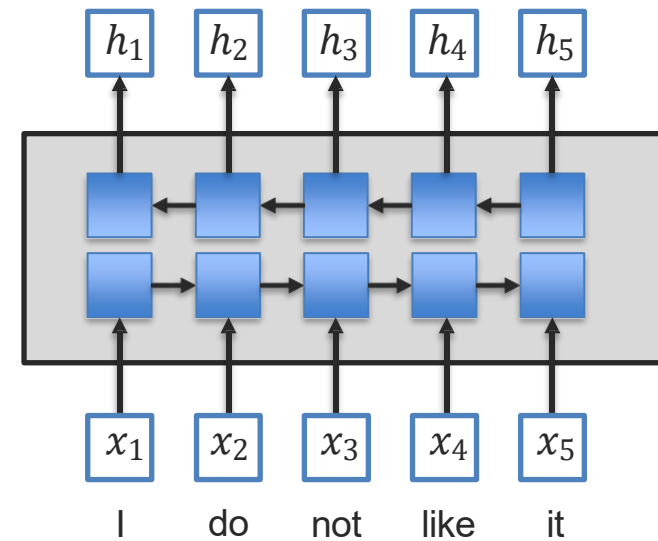
Contextualized Sequence Representations

Sequence Encoding - Contextualization



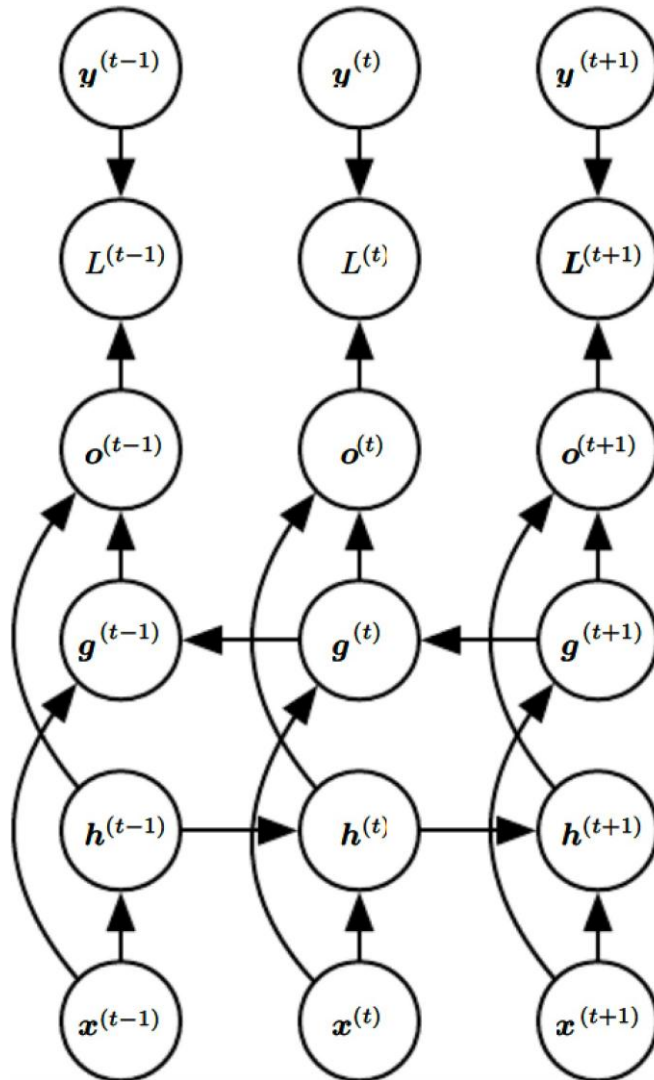
How to encode this sequence while modeling the interaction between elements (e.g., words)?

Option 1: Bi-directional LSTM:
(e.g., ELMO)



But harder to parallelize....

A typical bidirectional RNN



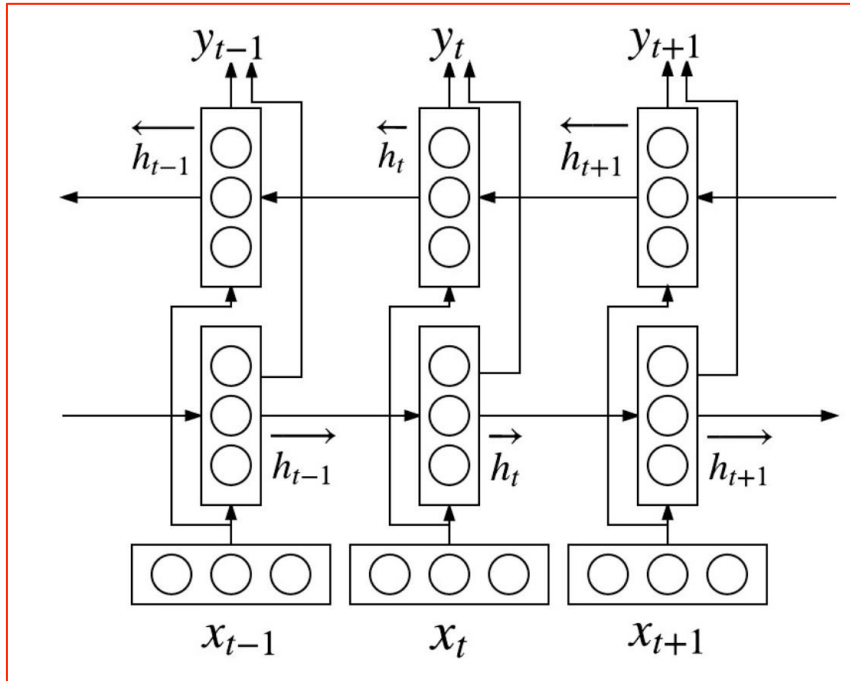
Maps input sequences x to target sequences y with loss $L^{(t)}$ at each step t

h recurrence propagates to the right
 g recurrence propagates to the left.

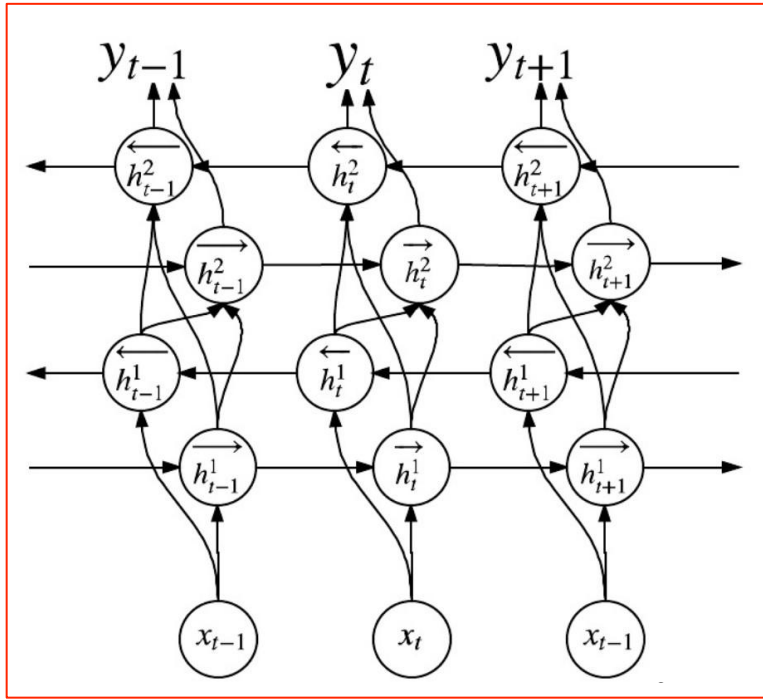
This allows output units $o^{(t)}$ to compute a representation that depends both the past and the future

Parameters of a Bidirectional RNN

Bidirectional RNN

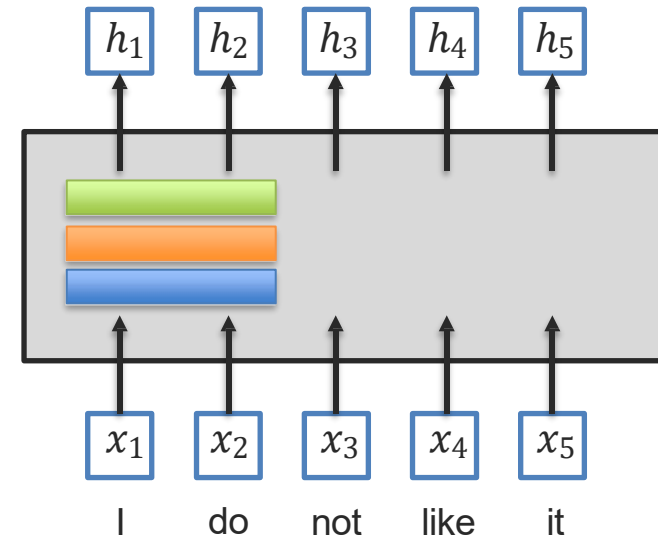
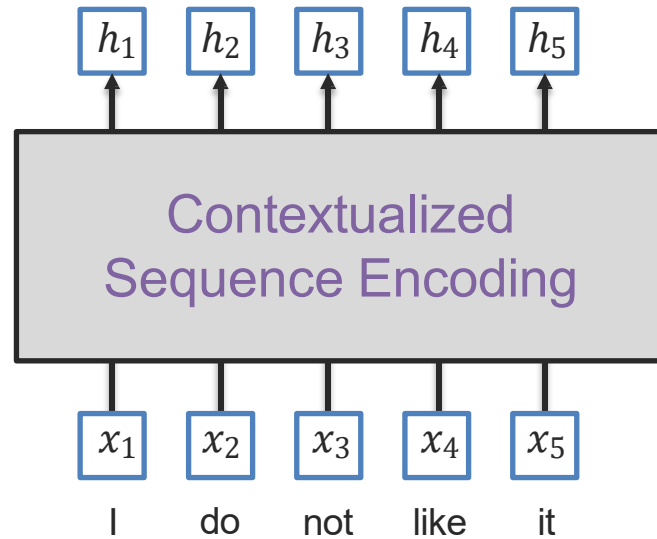


Deep Bidirectional RNN Has multiple layers per time step



Sequence Encoding - Contextualization

Option 2: Convolutions



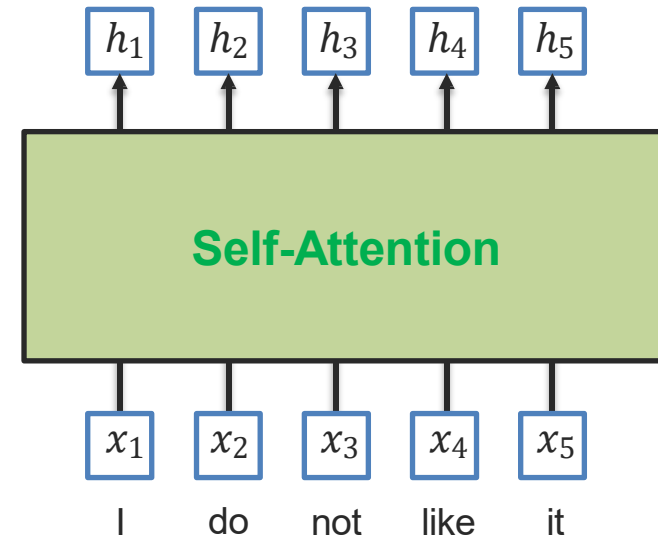
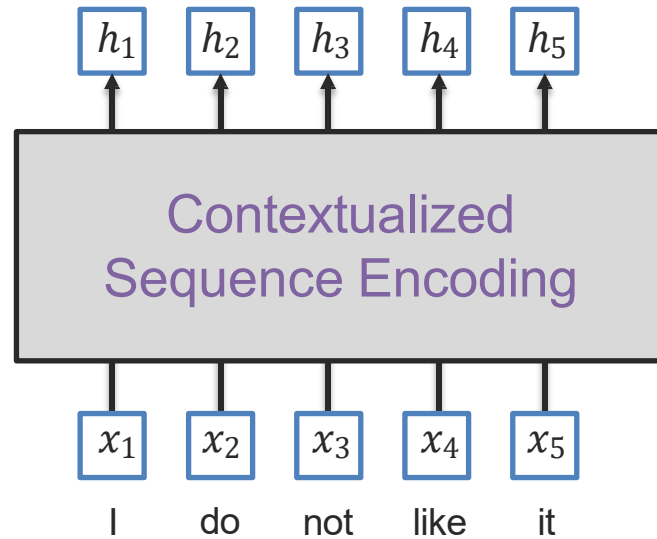
Can be parallelized!

But modeling long-range dependencies
require multiple layers

And convolutional kernels are static

Sequence Encoding - Contextualization

Option 3: Self-attention



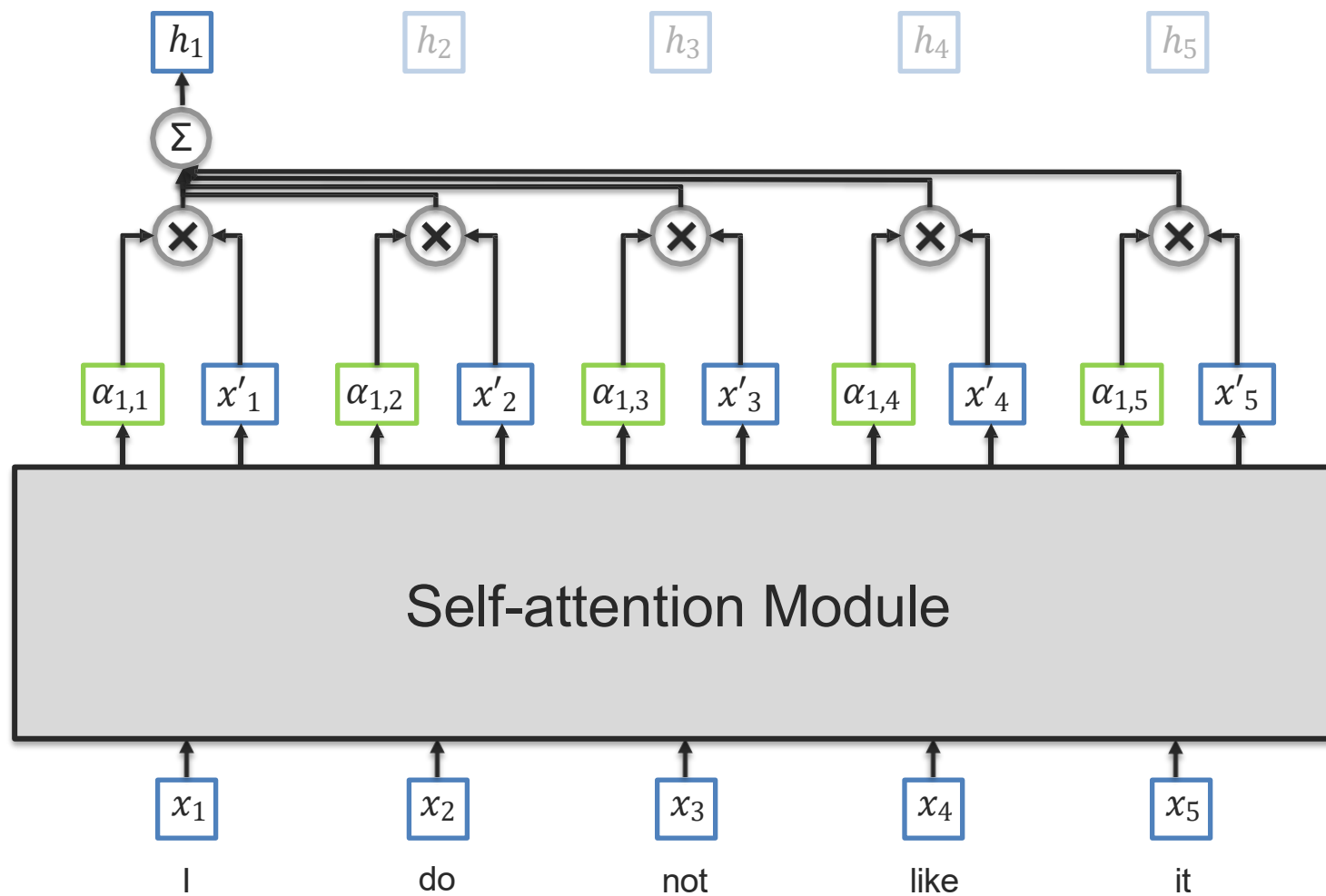
Can be parallelized!

Long-range dependencies

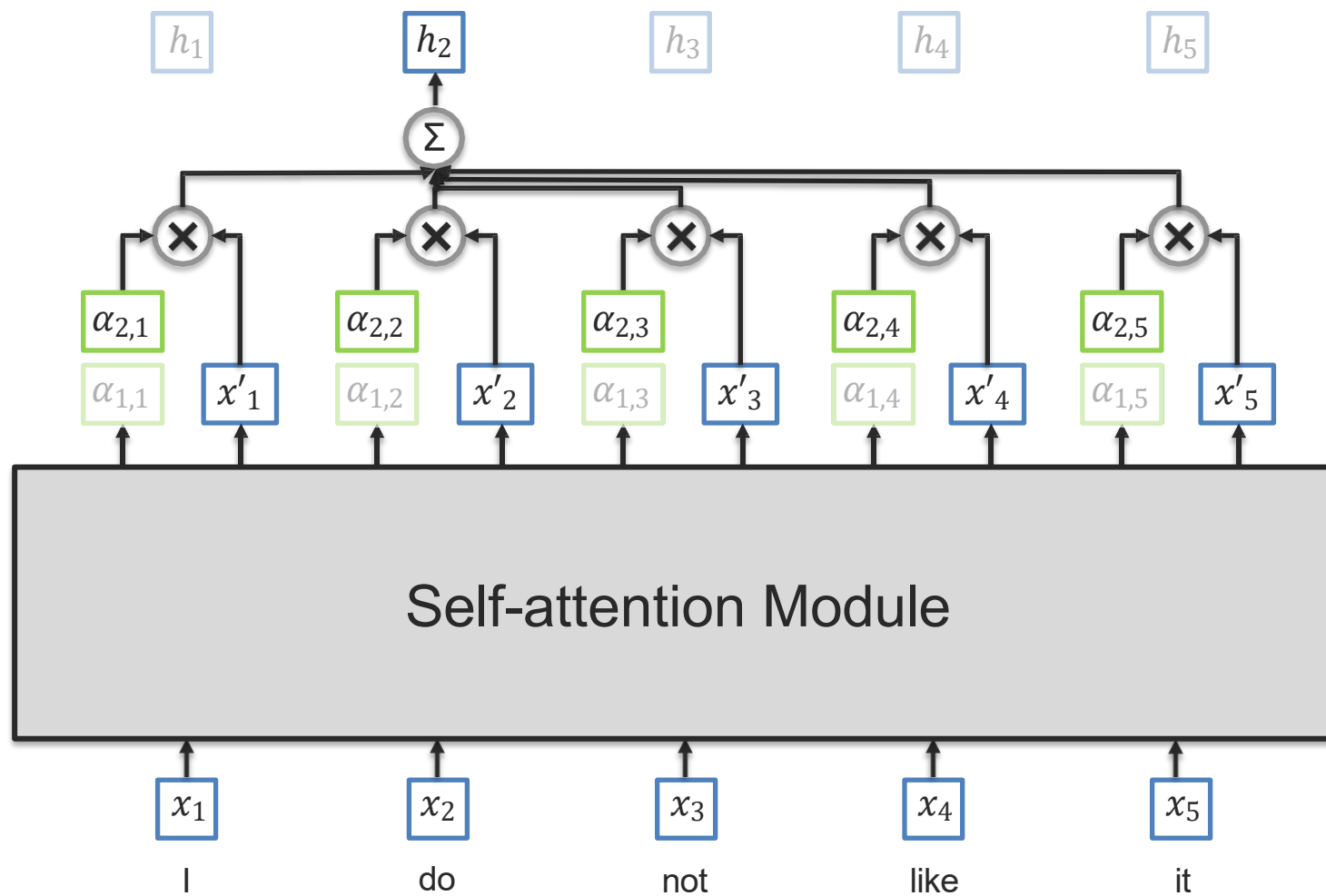
Dynamic attention weights

Self-Attention

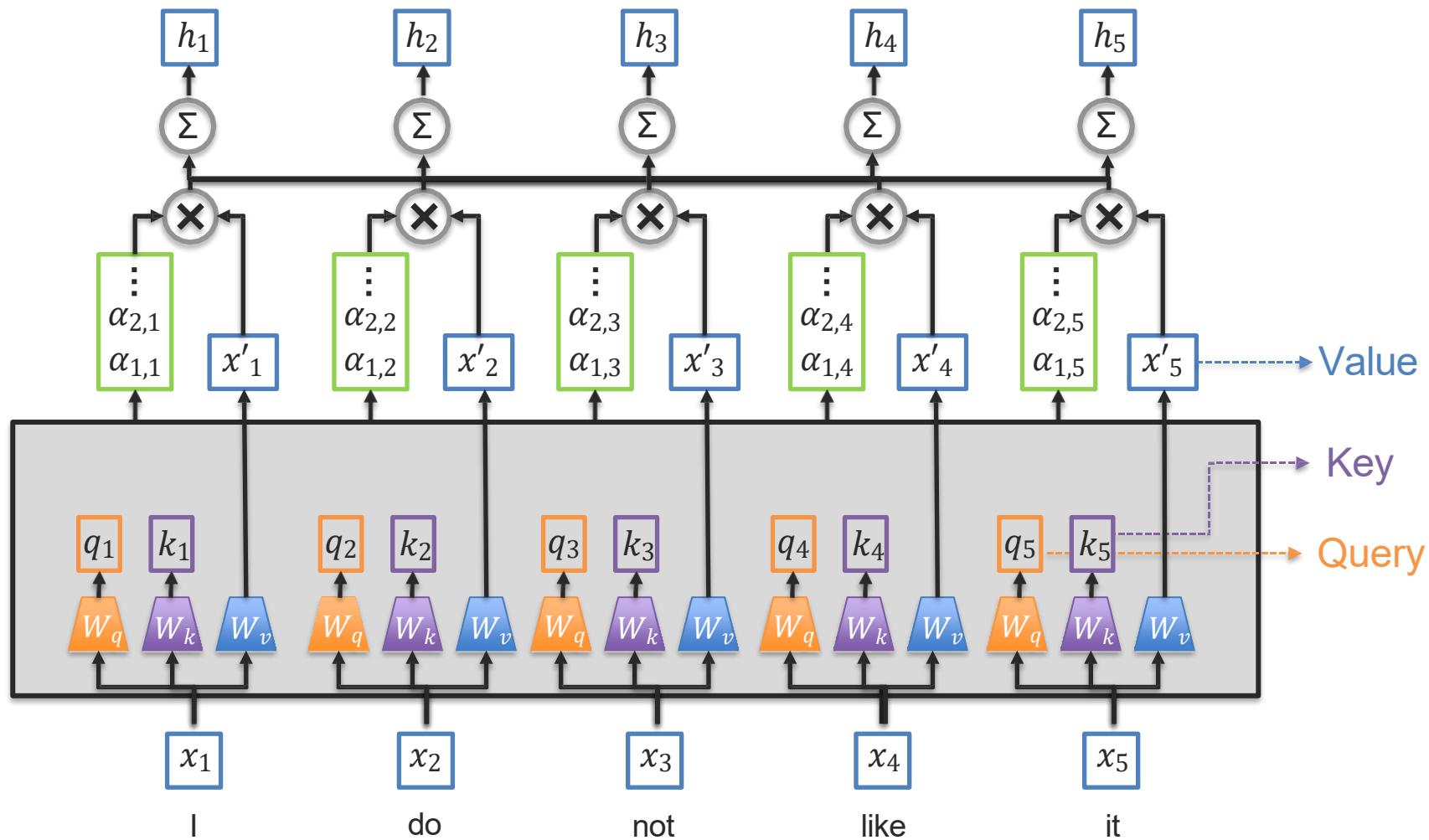
Self-Attention



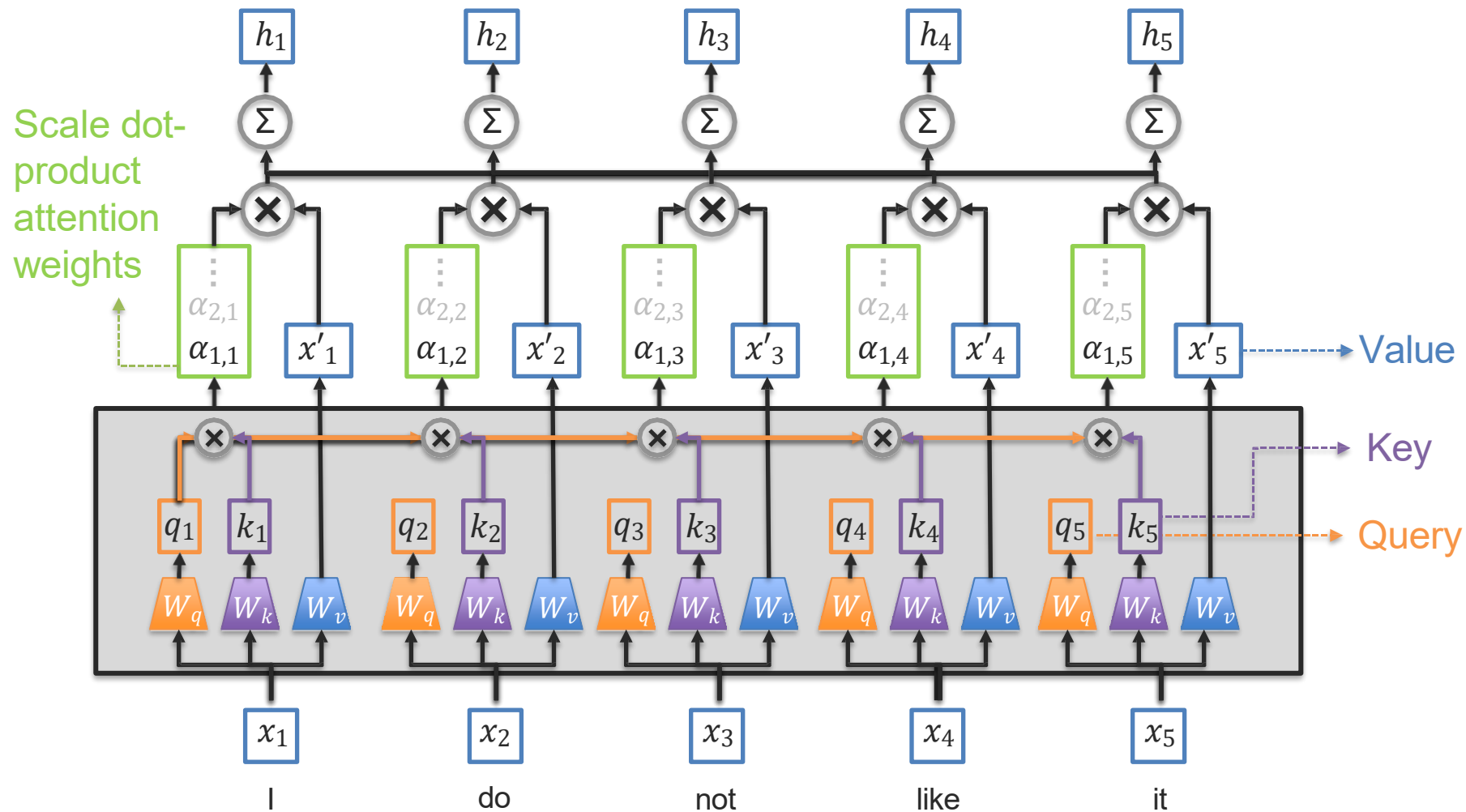
Self-Attention



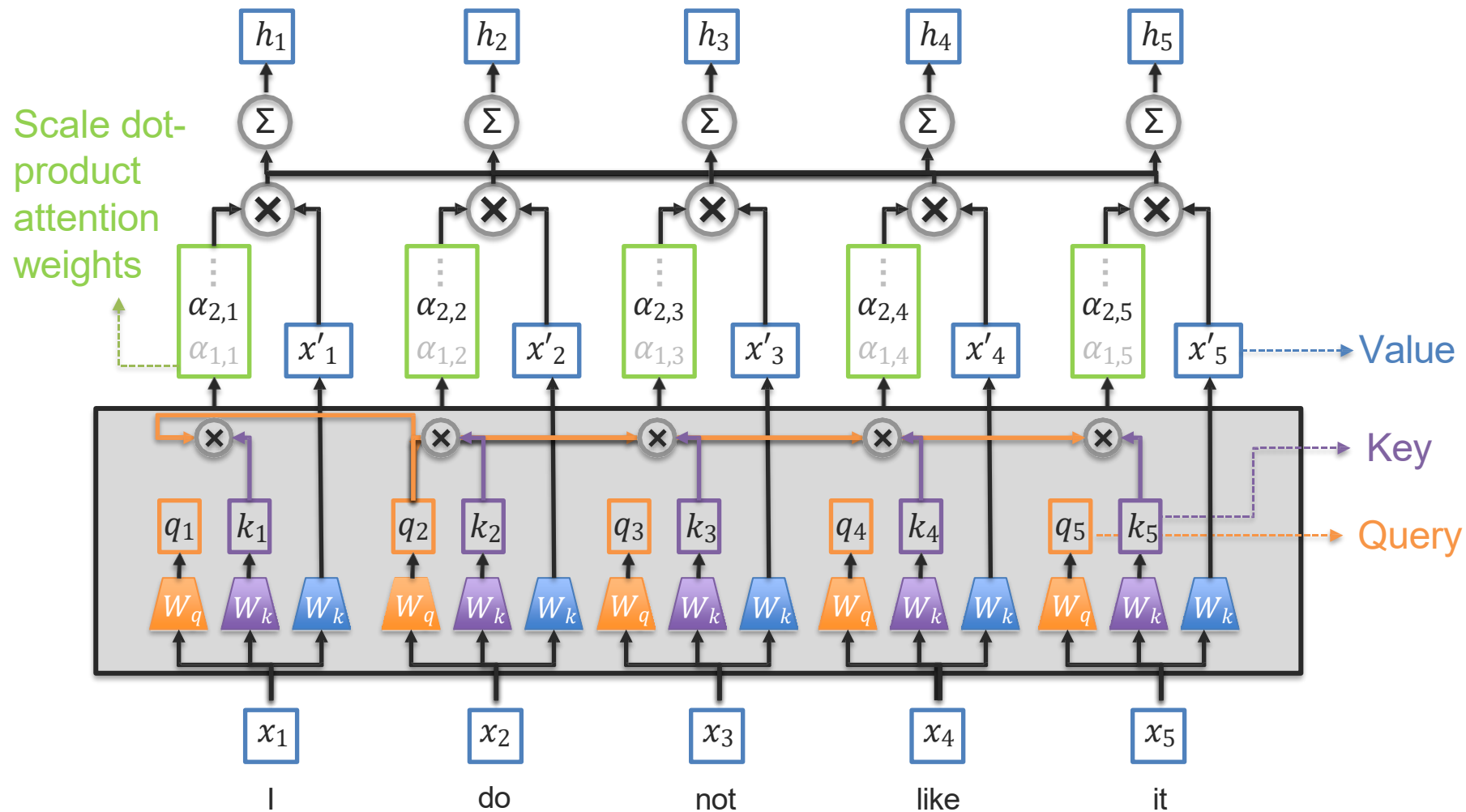
Transformer Self-Attention



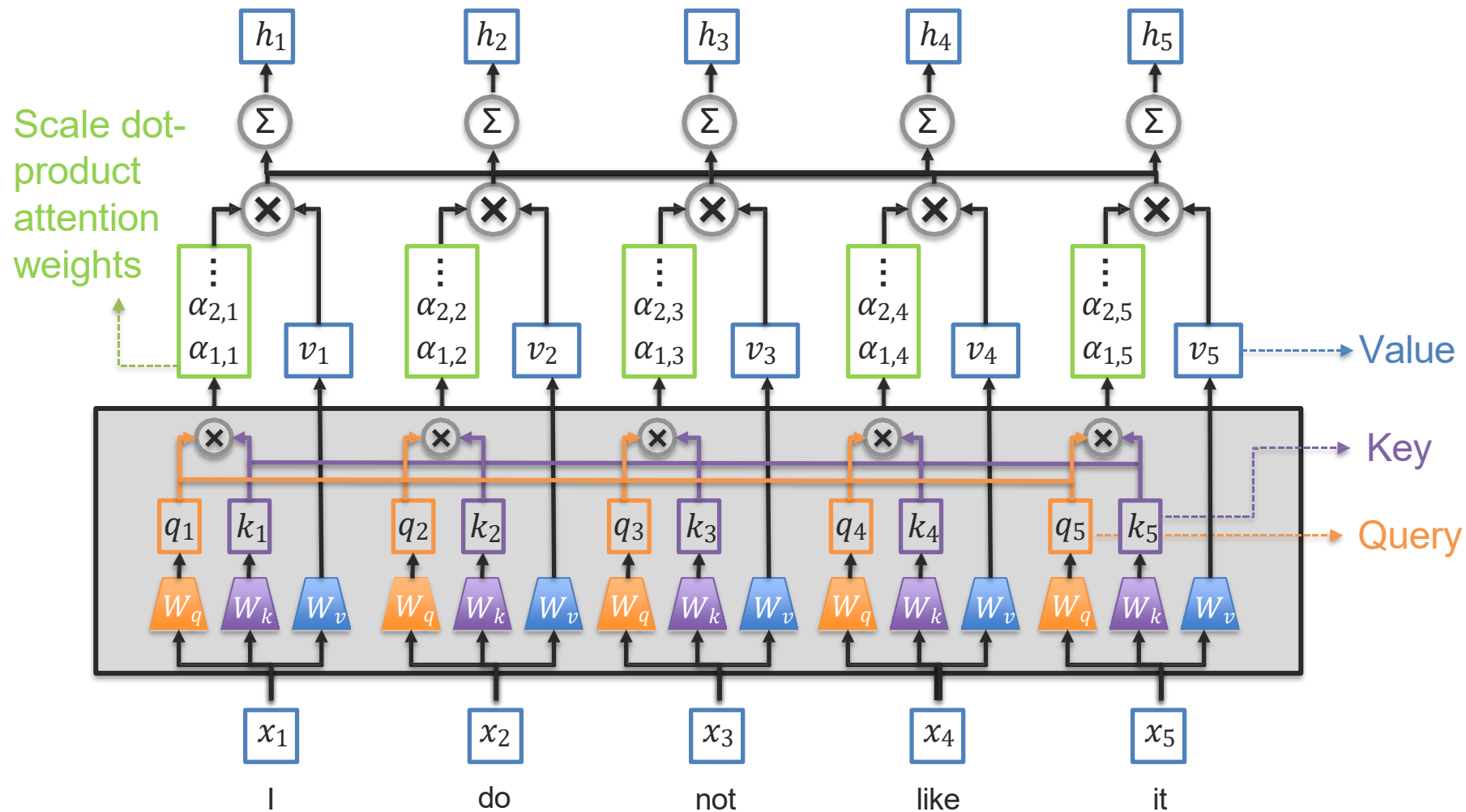
Transformer Self-Attention



Transformer Self-Attention

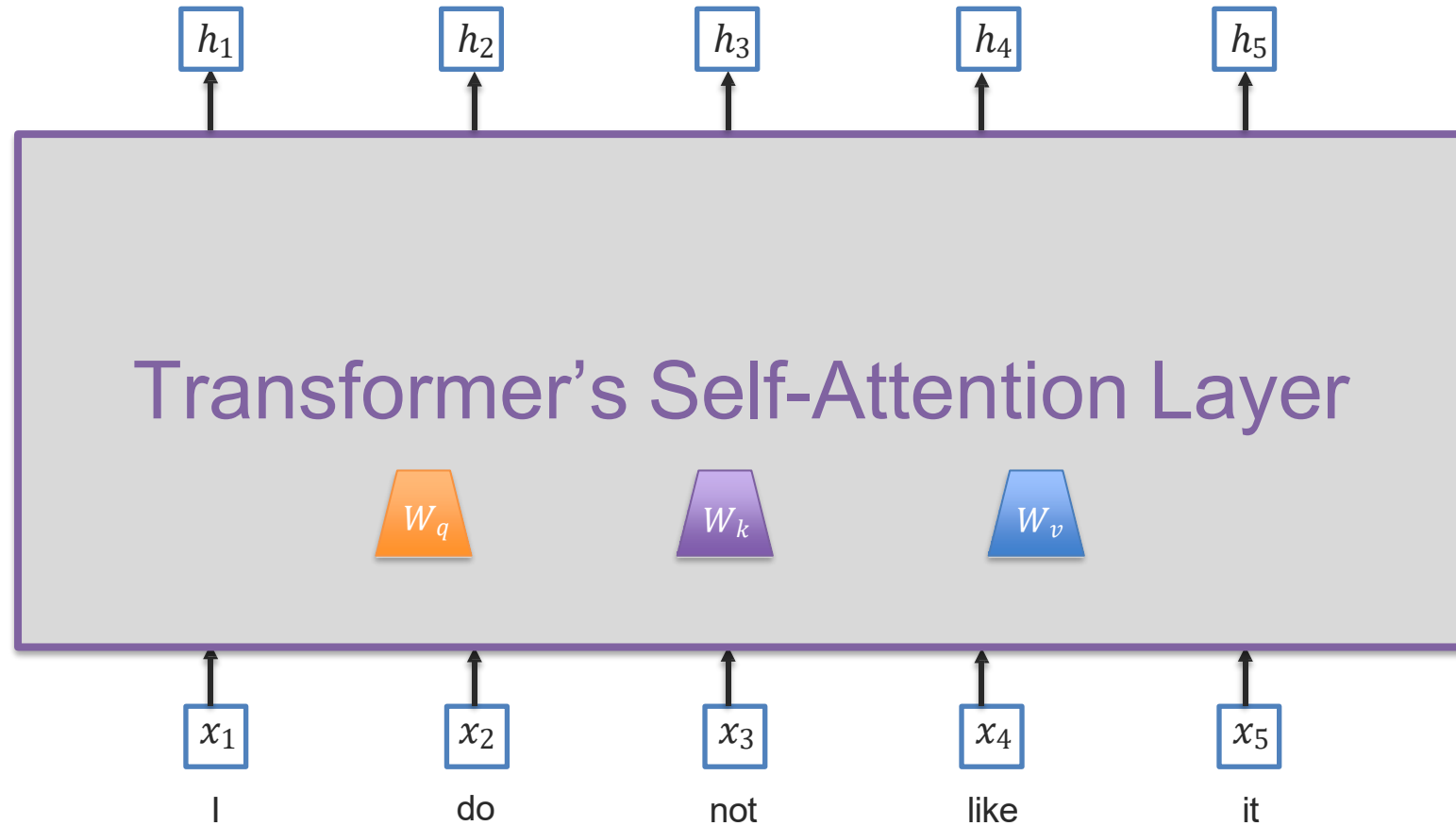


Transformer Self-Attention

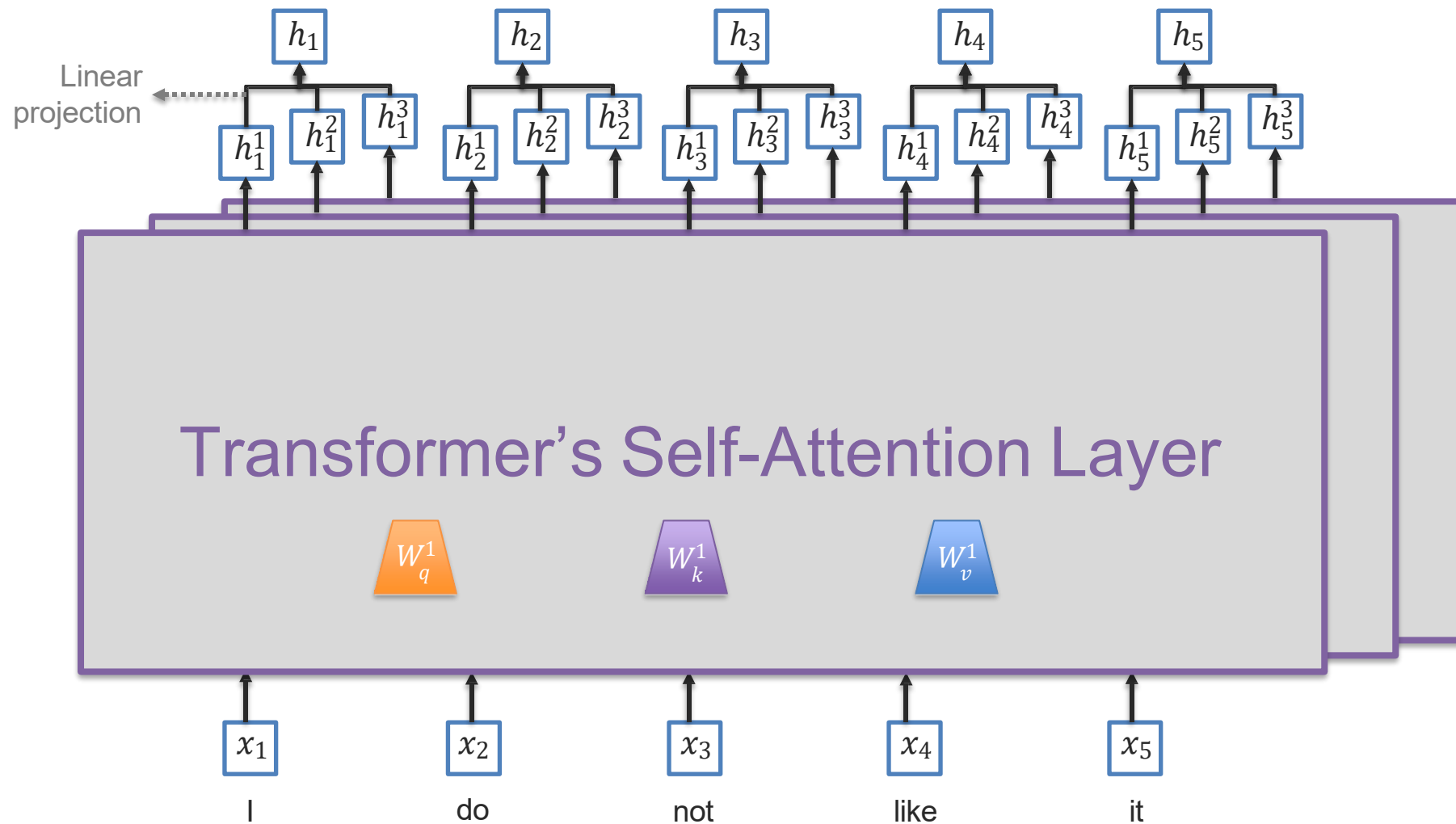


Transformer Self-Attention

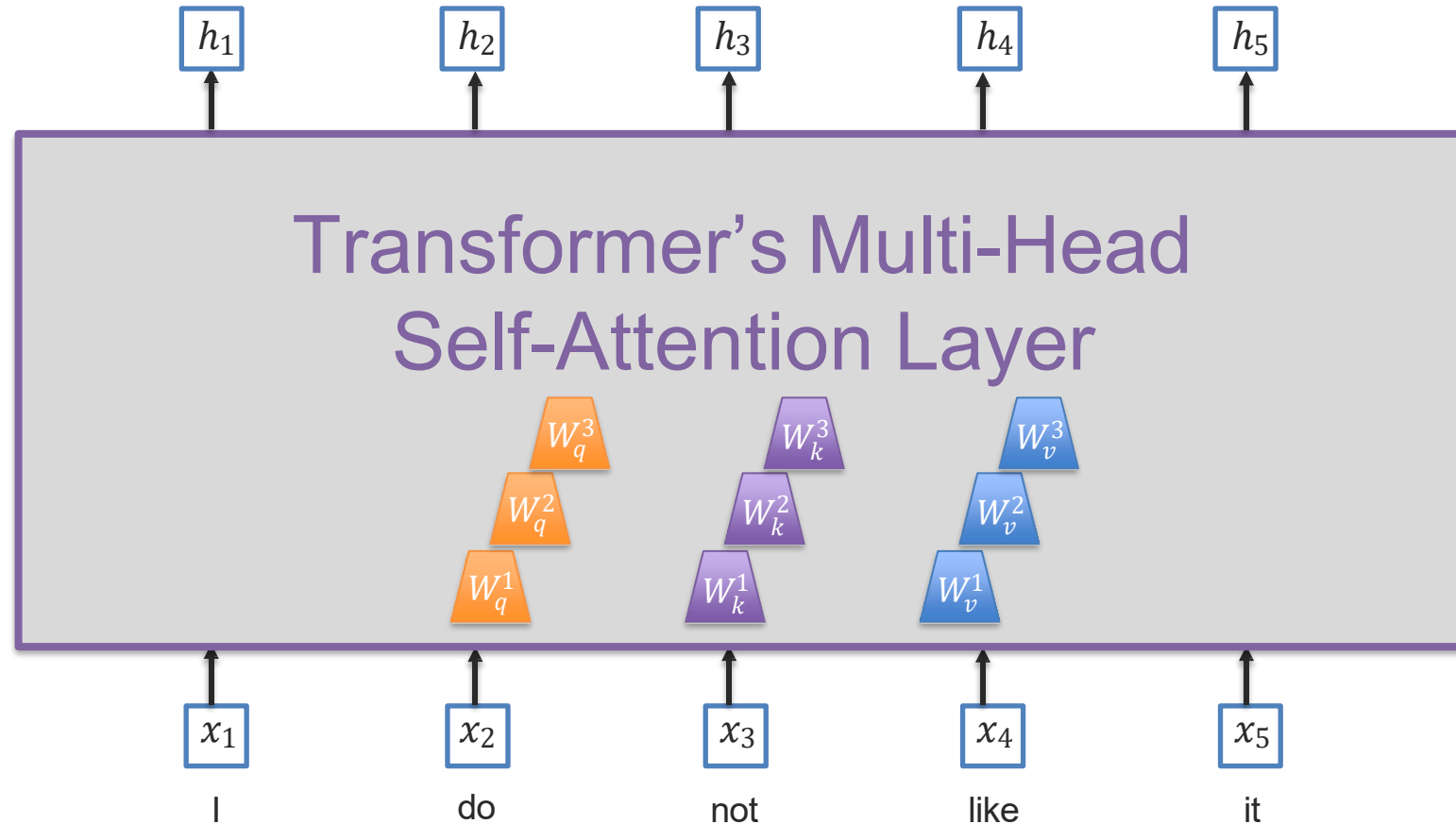
What if we want to attend simultaneously to multiple subspaces of x ?



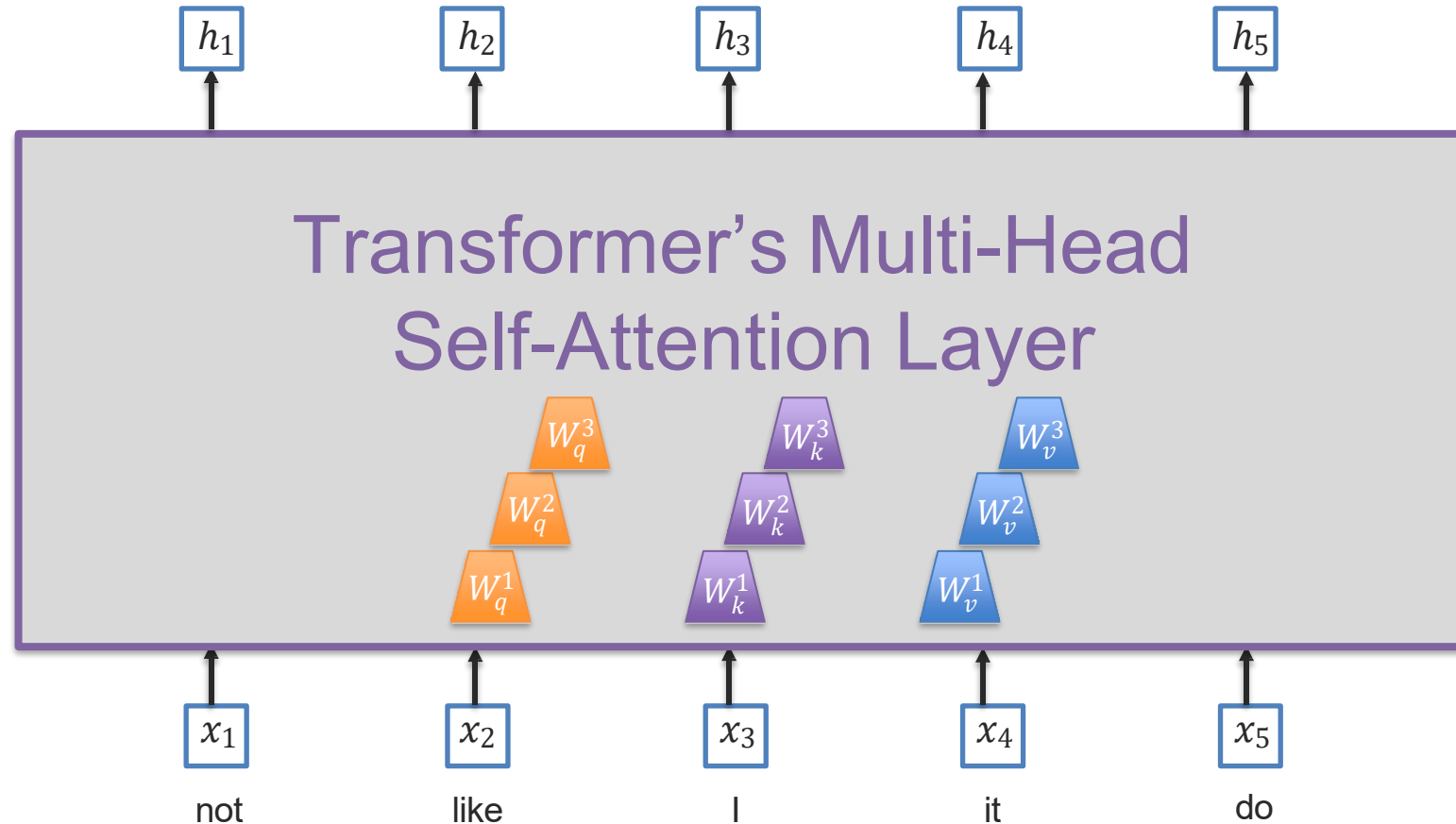
Transformer Multi-Head Self-Attention



Transformer Multi-Head Self-Attention



Transformer Multi-Head Self-Attention



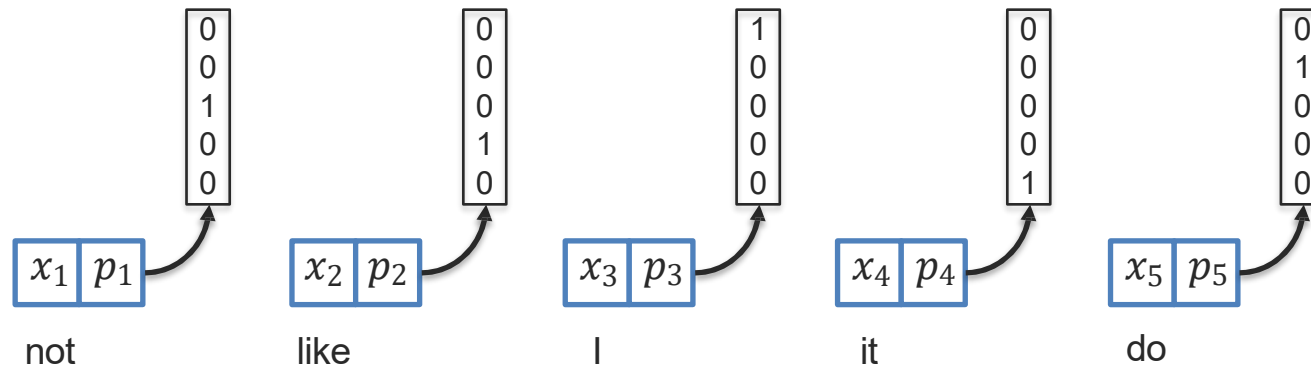
What happens if the words are shuffled?

Position embeddings

- ❑ Position information is not encoded in a self-attention module

How can we encode position information?

Simple approach: one-hot encoding

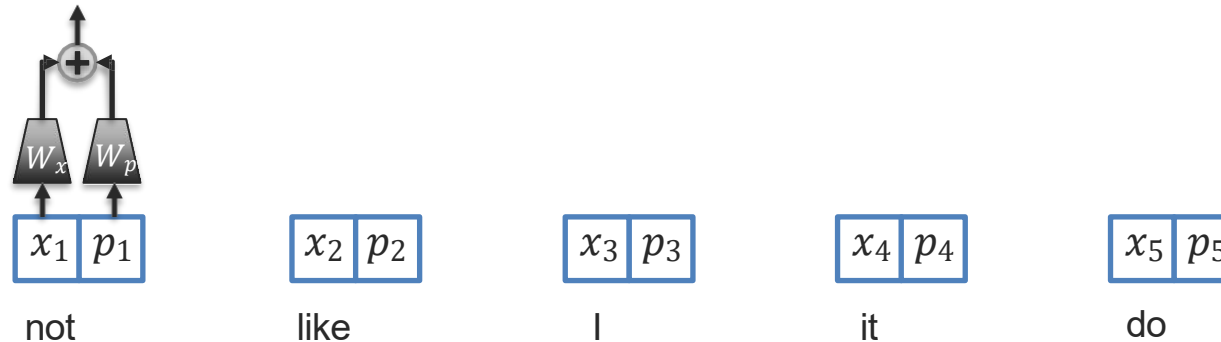


Position embeddings

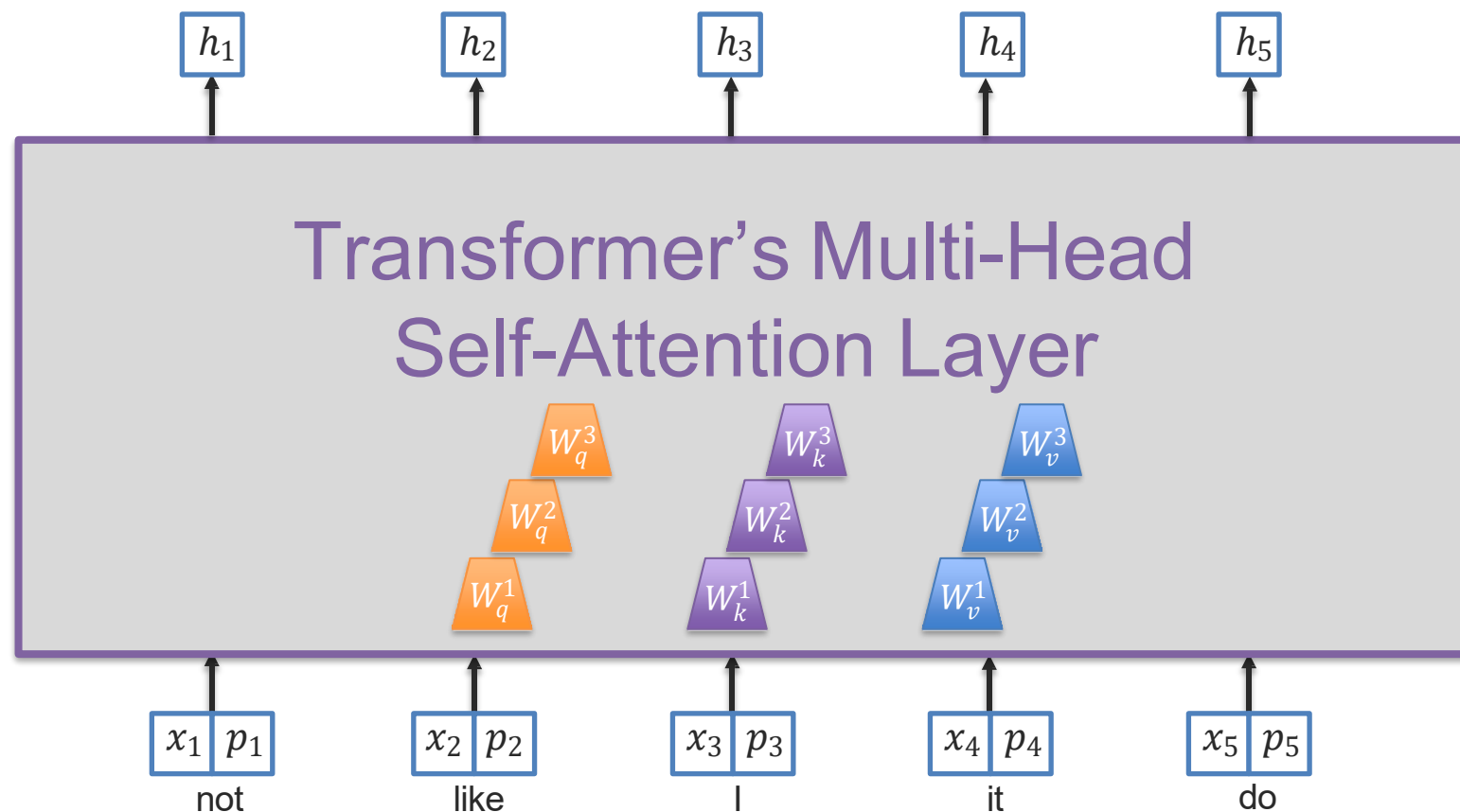
- ❑ Position information is not encoded in a self-attention module

How can we encode position information?

Simple approach: one-hot encoding + linear embeddings + $\left\{ \begin{array}{l} \text{Sum} \\ \text{- or -} \\ \text{concat} \end{array} \right.$

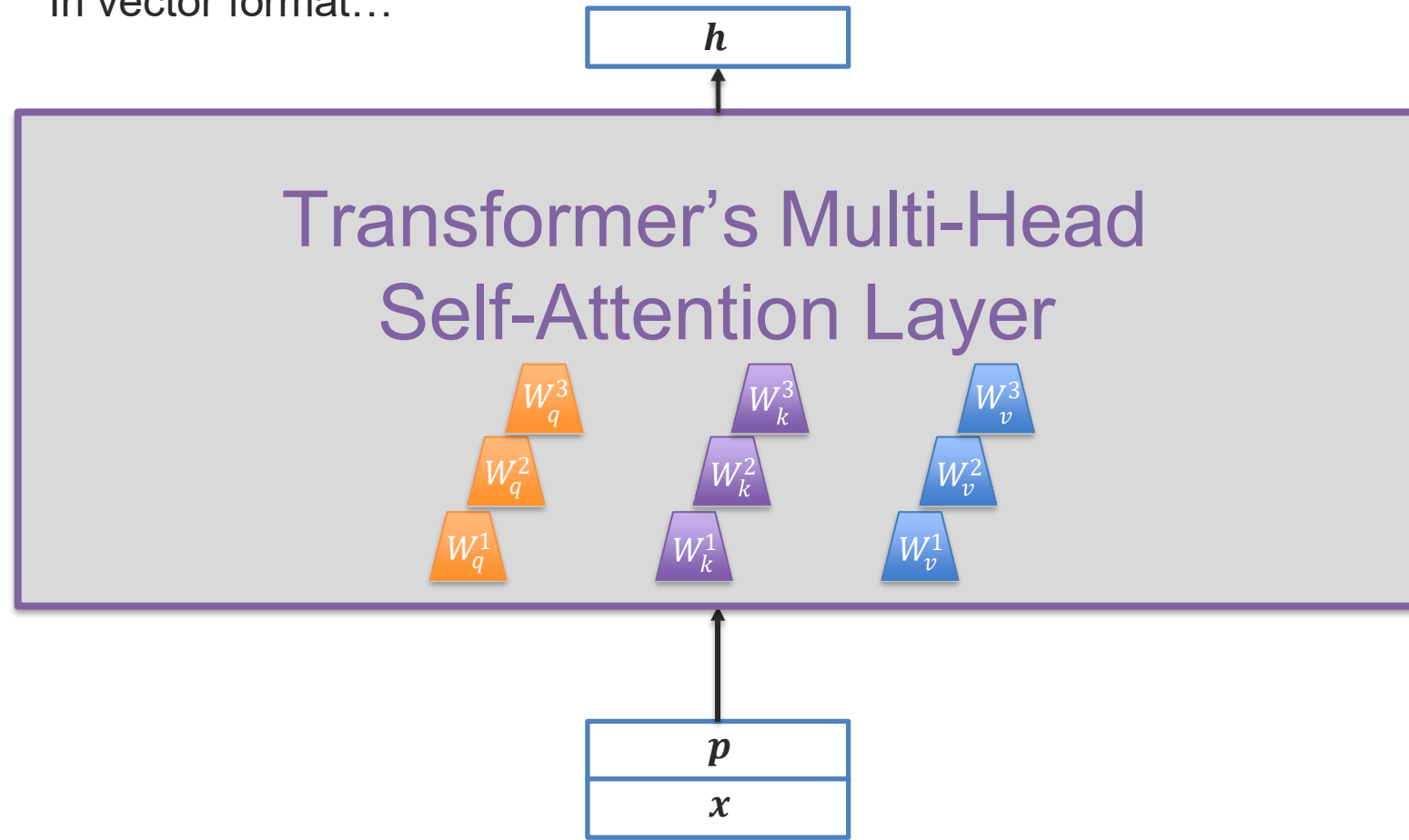


Transformer Multi-Head Self-Attention

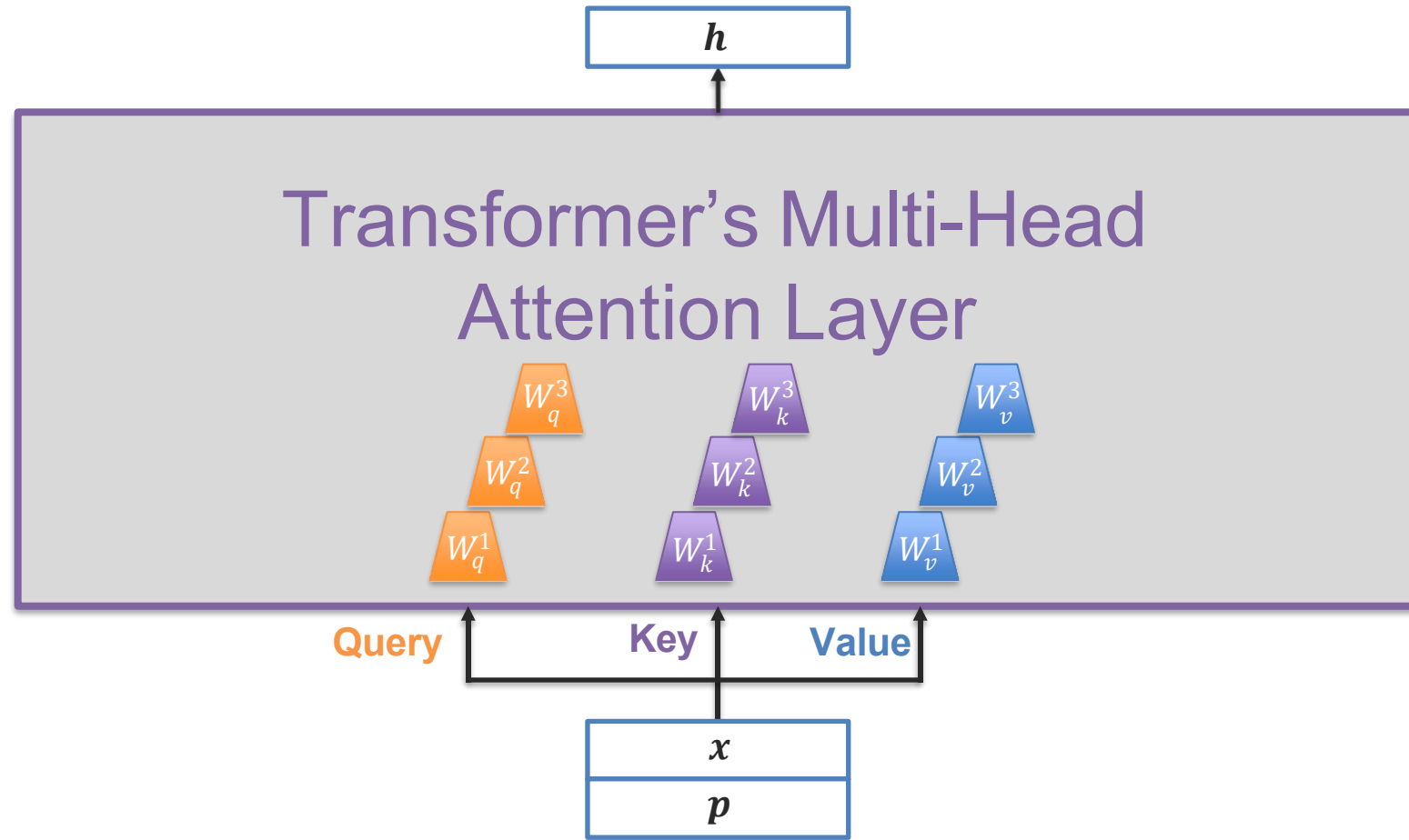


Transformer Multi-Head Self-Attention

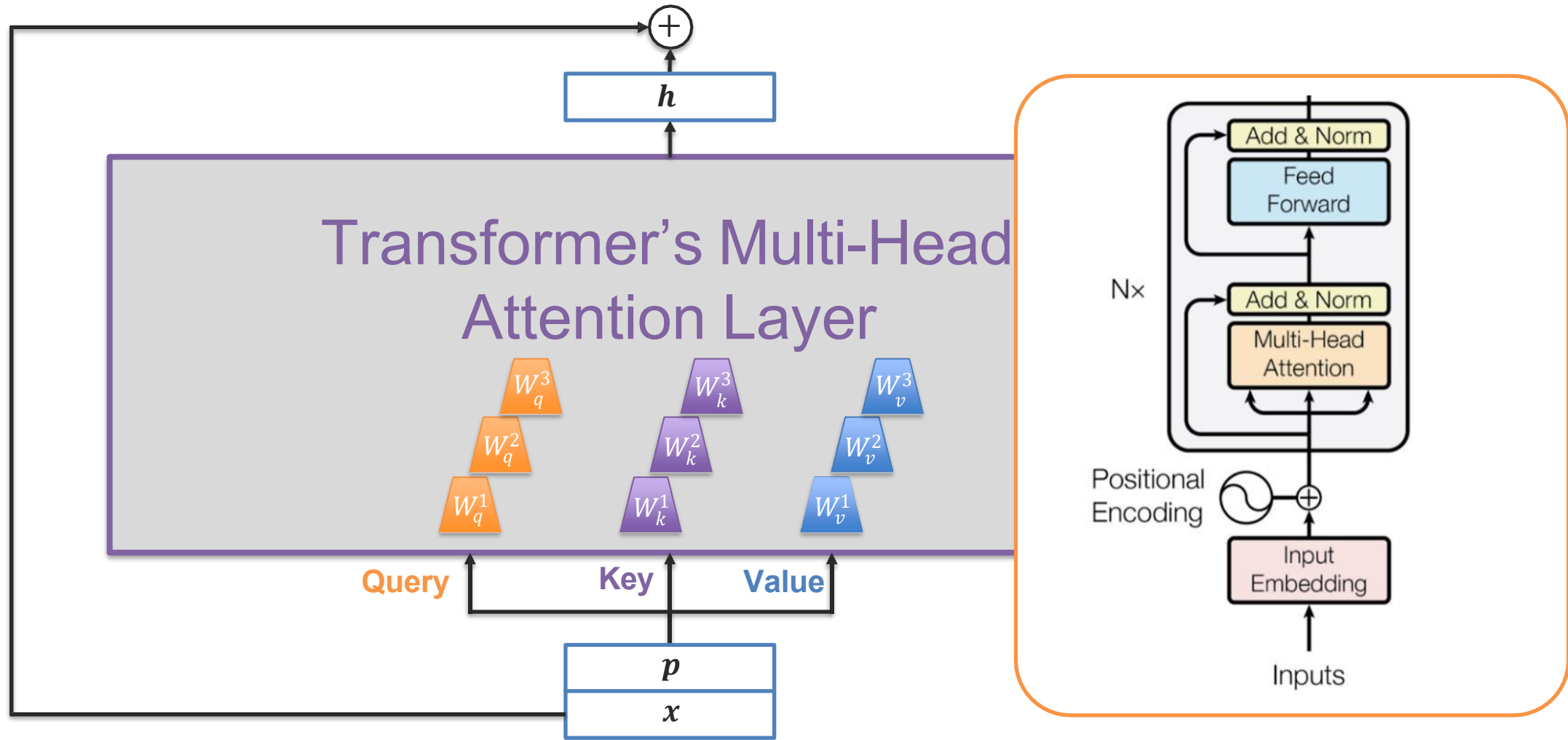
In vector format...



Transformer Multi-Head Attention



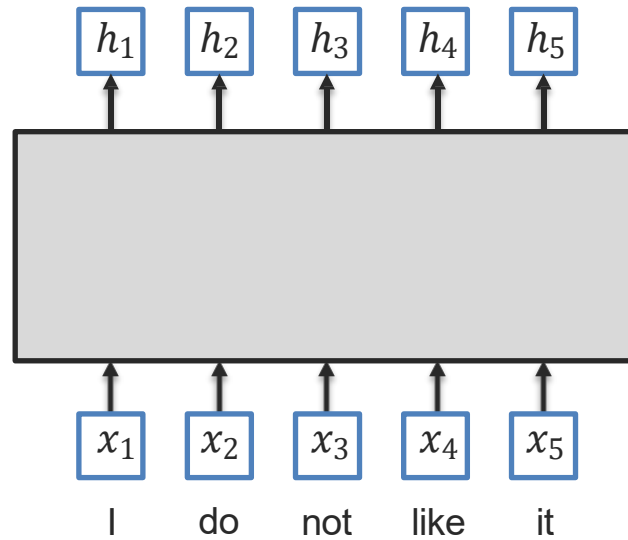
Transformer – Residual Connection



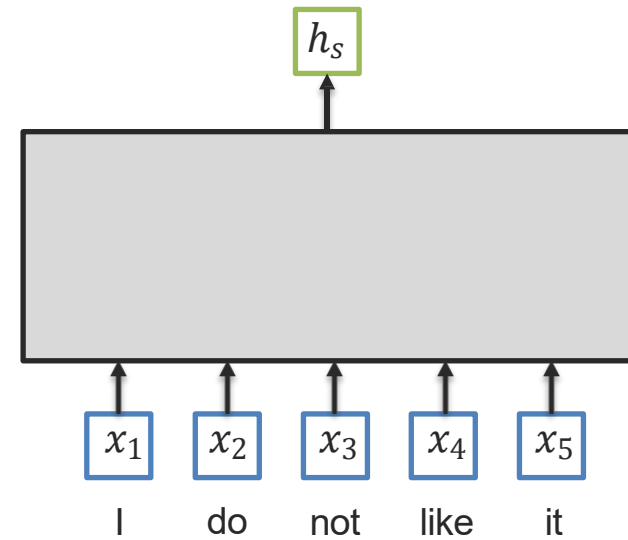
Language Pre-training

Token-level and Sentence-level Embeddings

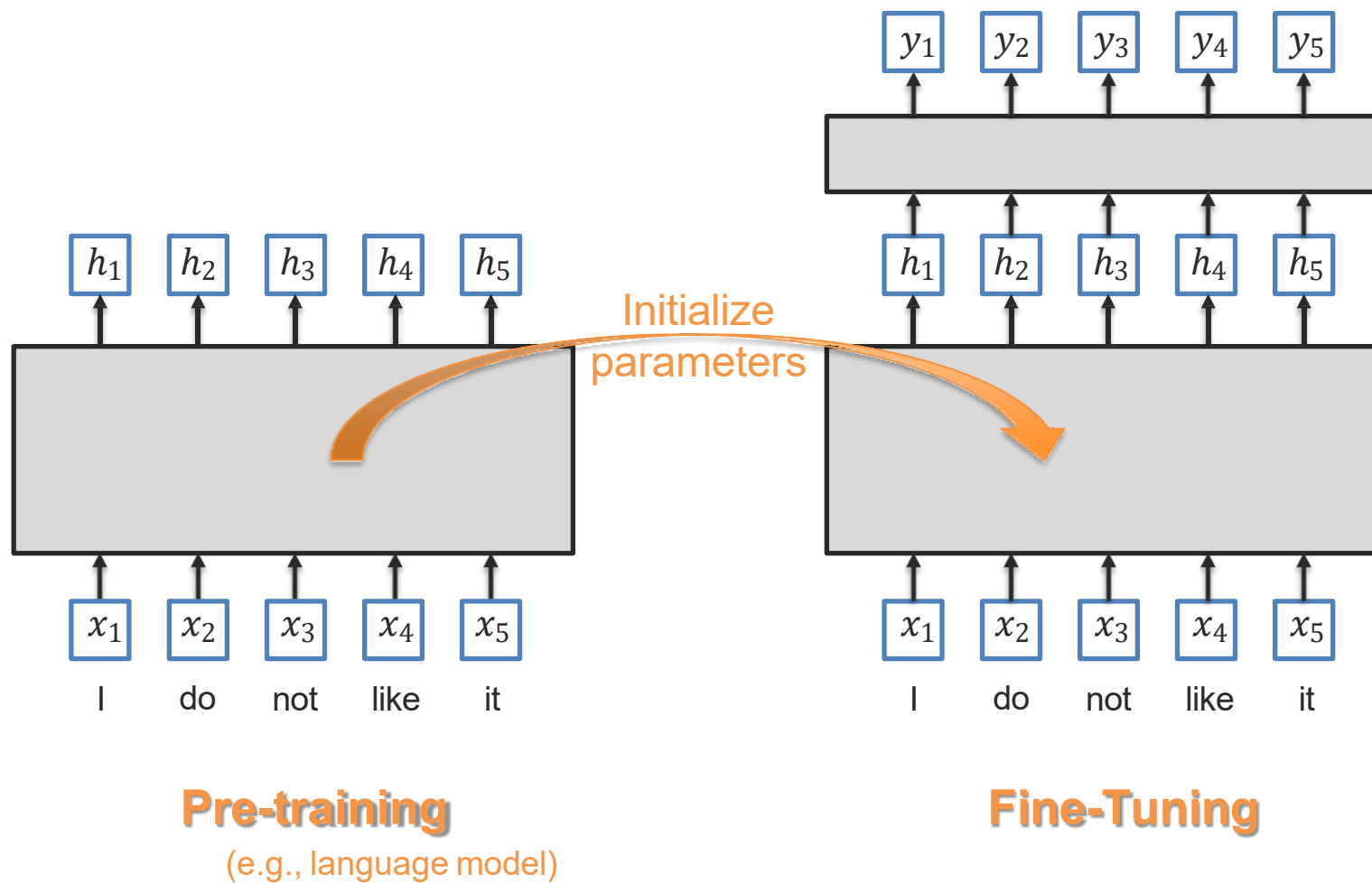
Token-level embeddings



Sentence-level embedding



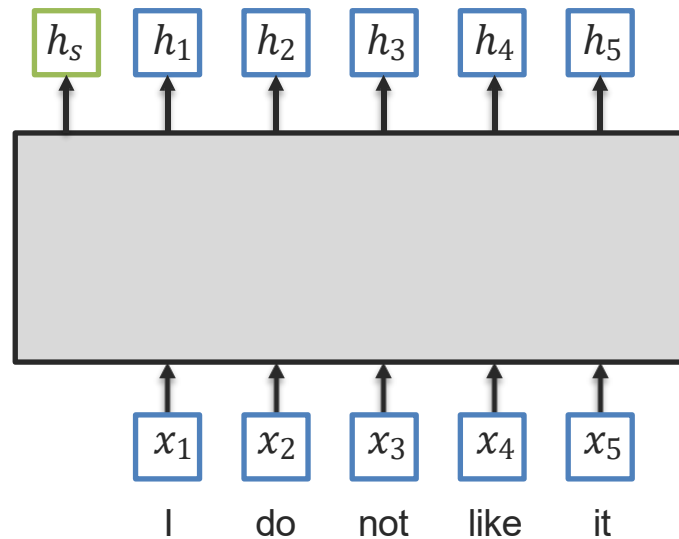
Pre-Training and Fine-Tuning



BERT: Bidirectional Encoder Representations from Transformers

Advantages:

- ① Jointly learn representation for token-level and sentence level
- ② Same network architecture for pre-training and fine-tuning

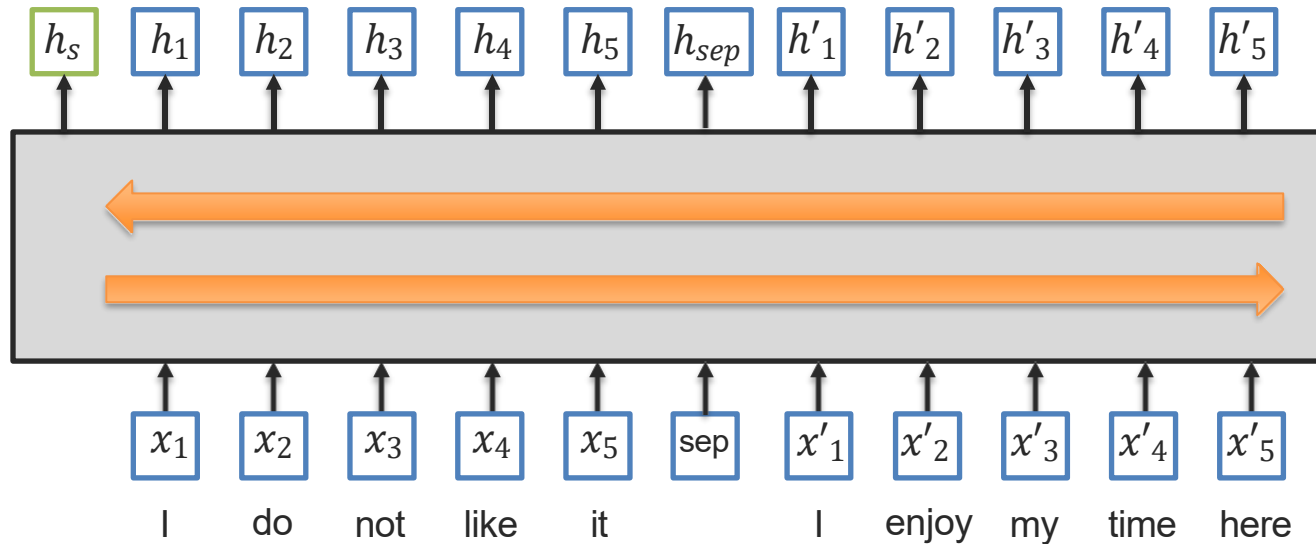


BERT: Bidirectional Encoder Representations from Transformers

Advantages:

- 1 Jointly learn representation for token-level and sentence level
- 2 Same network architecture for pre-training and fine-tuning
- 3 Can be used learn relationship between sentences
- 4 Models bidirectional and long-range interactions between tokens

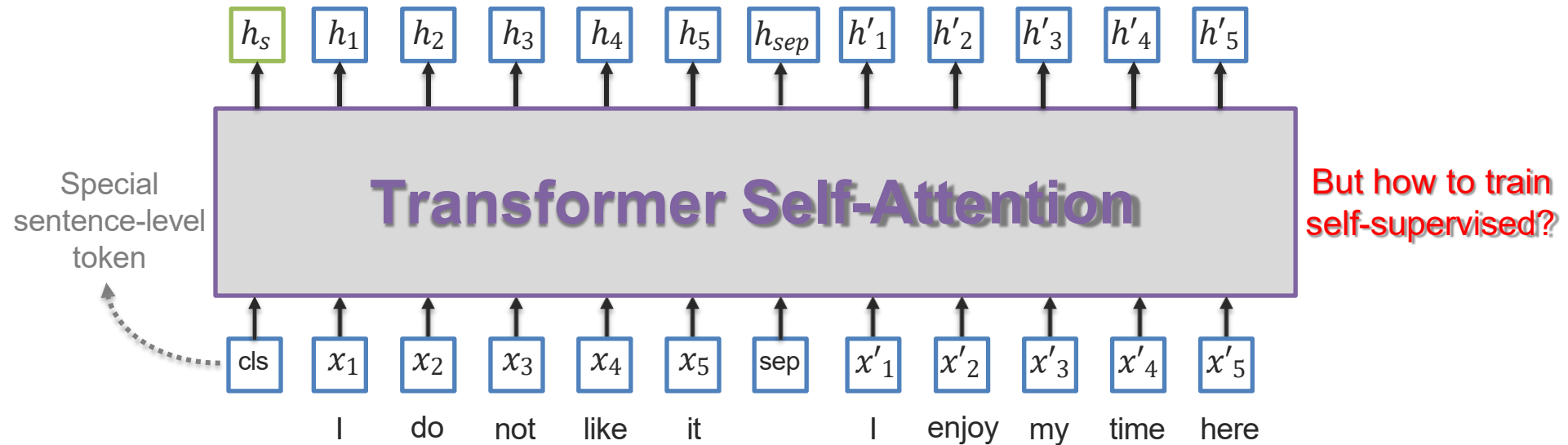
How can
we do all
this?



BERT: Bidirectional Encoder Representations from Transformers

Advantages:

- ① Jointly learn representation for token-level and sentence level
- ② Same network architecture for pre-training and fine-tuning
- ③ Can be used learn relationship between sentences
- ④ Models bidirectional interactions between tokens

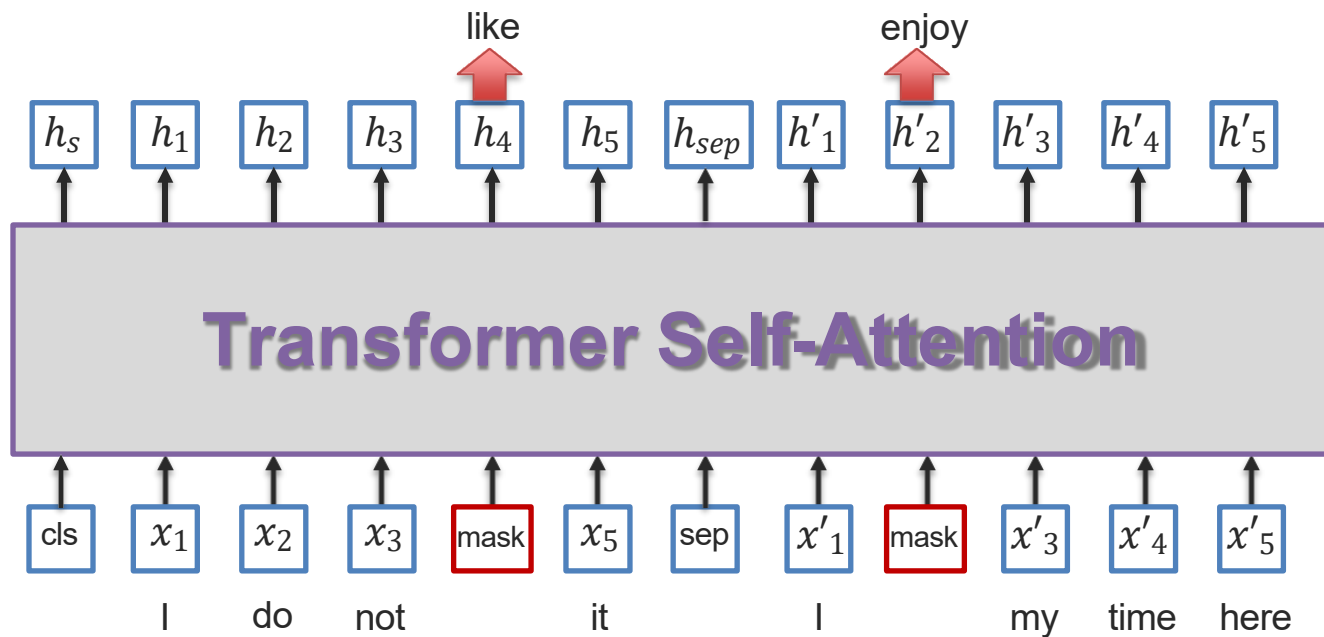


Pre-training BERT Model

1 Masked Language Model

Randomly mask input tokens and then try to predict them

What is the loss function?



Pre-training BERT Model

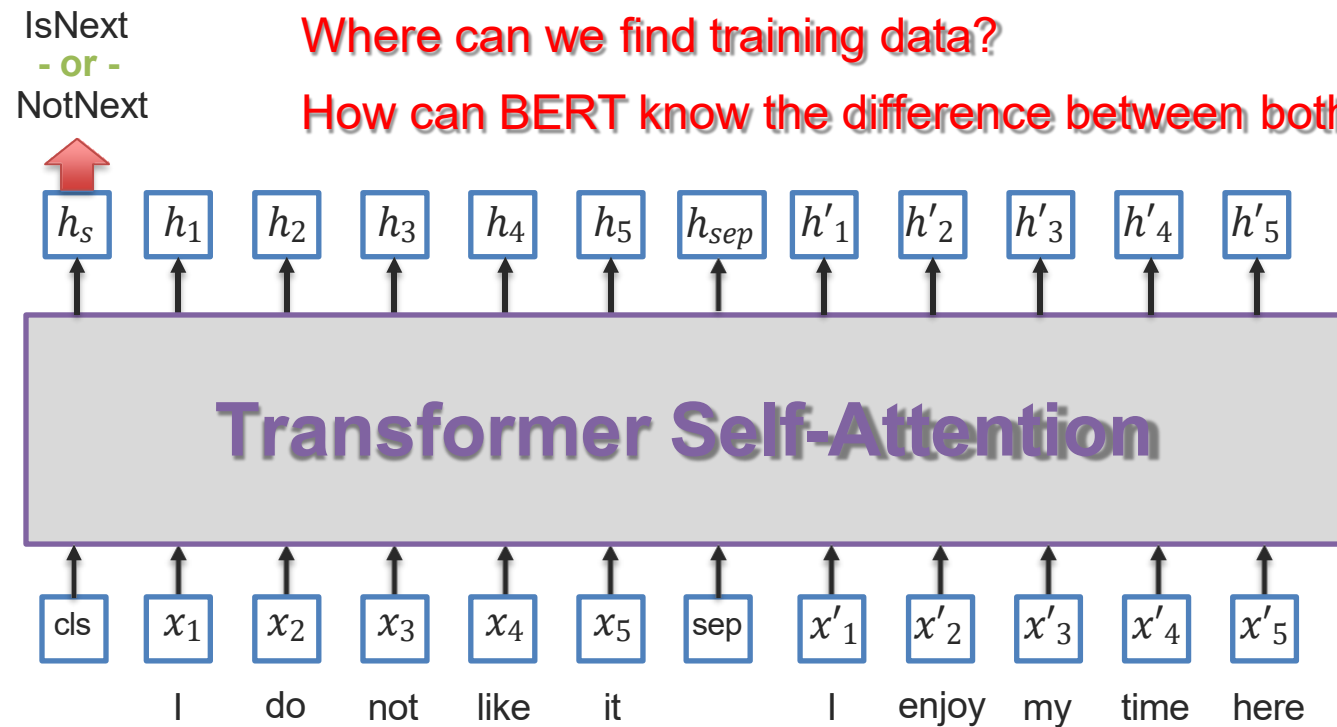
② Next Sentence Prediction

Given two sentences, predict if this is the next one or not

What is the loss function?

Where can we find training data?

How can BERT know the difference between both sentences?



Next Sentence Prediction (NSP)

IsNext ✓

Sentence A:

[CLS] The man went to the store

Sentence B:

[SEP] He bought milk [SEP]

$h_1 ([CLS]) \rightarrow \text{Linear} \rightarrow \text{IsNext}$

Consecutive sentences from corpus

NotNext ✗

Sentence A:

[CLS] The man went to the store

Sentence B:

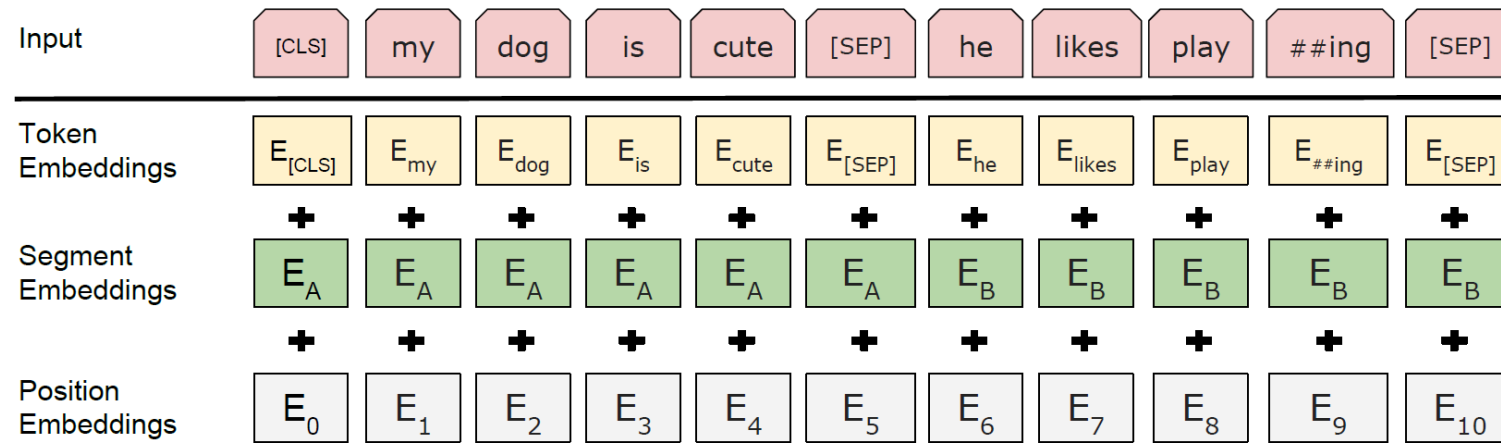
[SEP] Cats sleep a lot [SEP]

$h_1 ([CLS]) \rightarrow \text{Linear} \rightarrow \text{NotNext}$

Random sentence from corpus

NSP also uses a Linear head on h_1 (the [CLS] vector), NOT the raw hidden state directly

Three Embeddings: Token + Position + Sentence

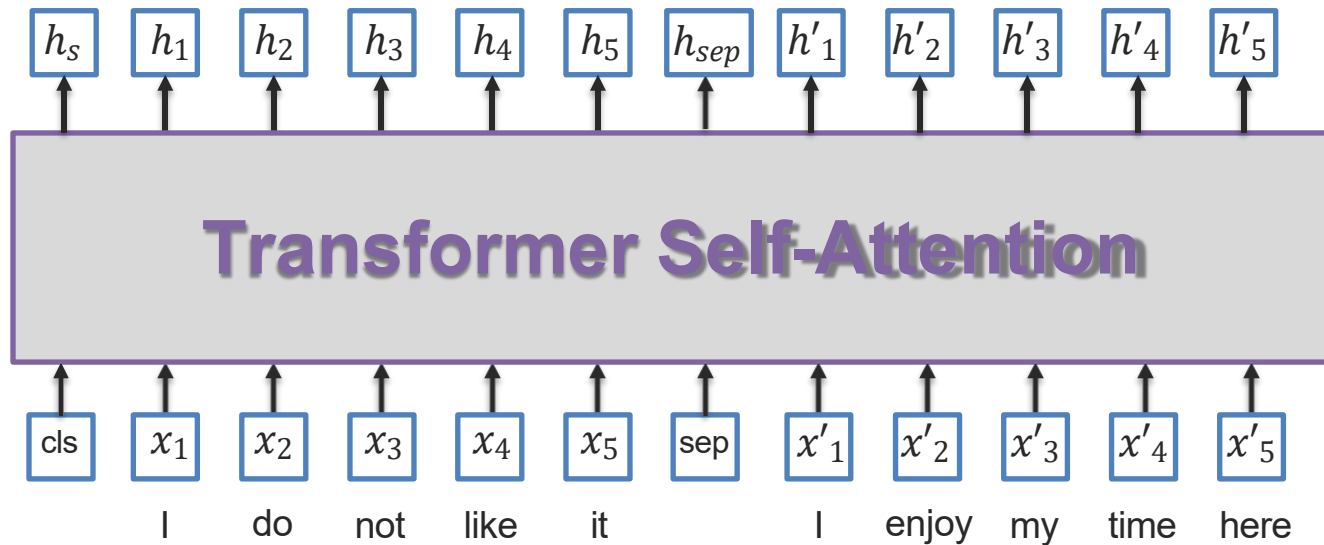


Fine-Tuning BERT

- 1 Sentence-level classification for only one sentence

Examples: sentiment analysis, document classification

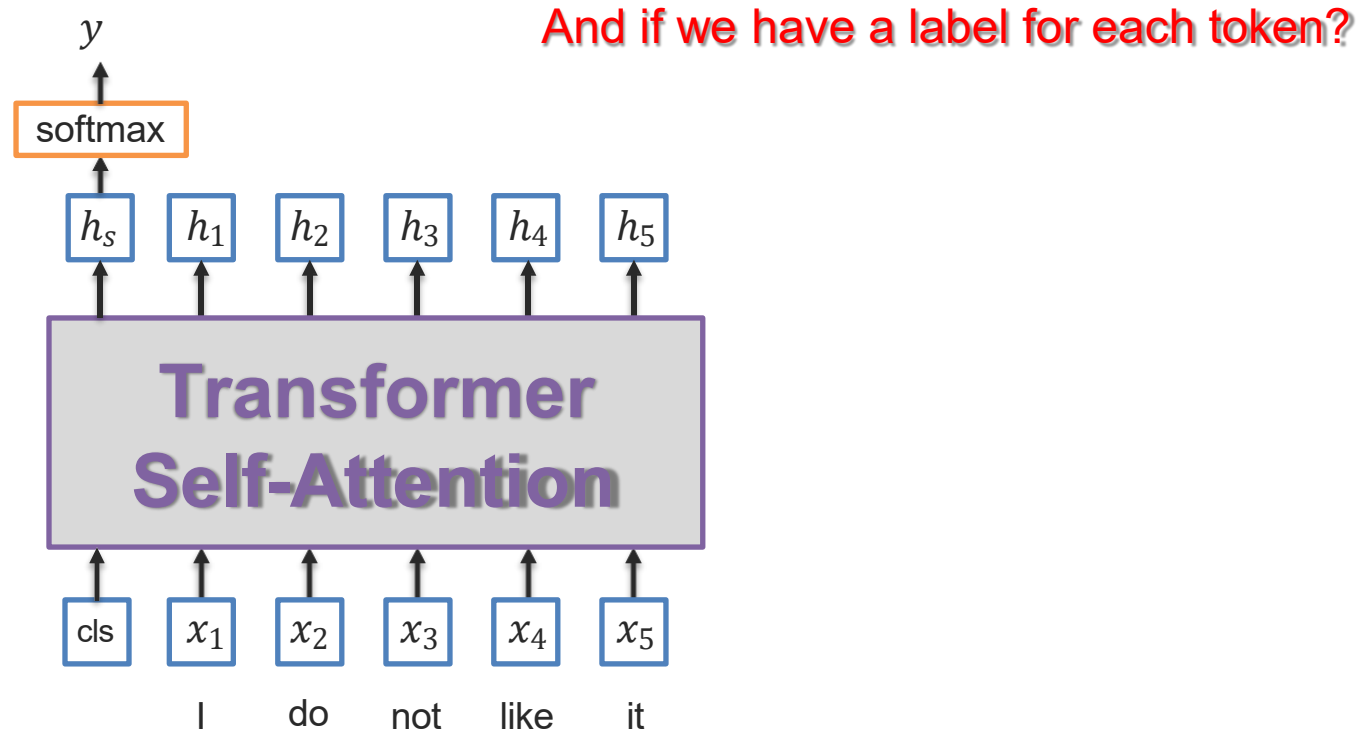
How?



Fine-Tuning BERT

- 1 Sentence-level classification for only one sentence

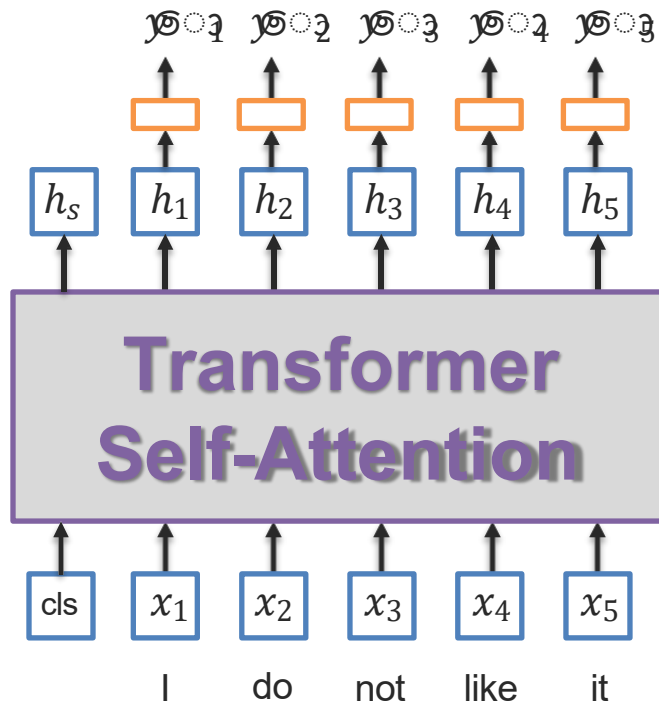
Examples: sentiment analysis, document classification



Fine-Tuning BERT

- ② Token-level classification for only one sentence

Examples: part-of-speech tagging, slot filling

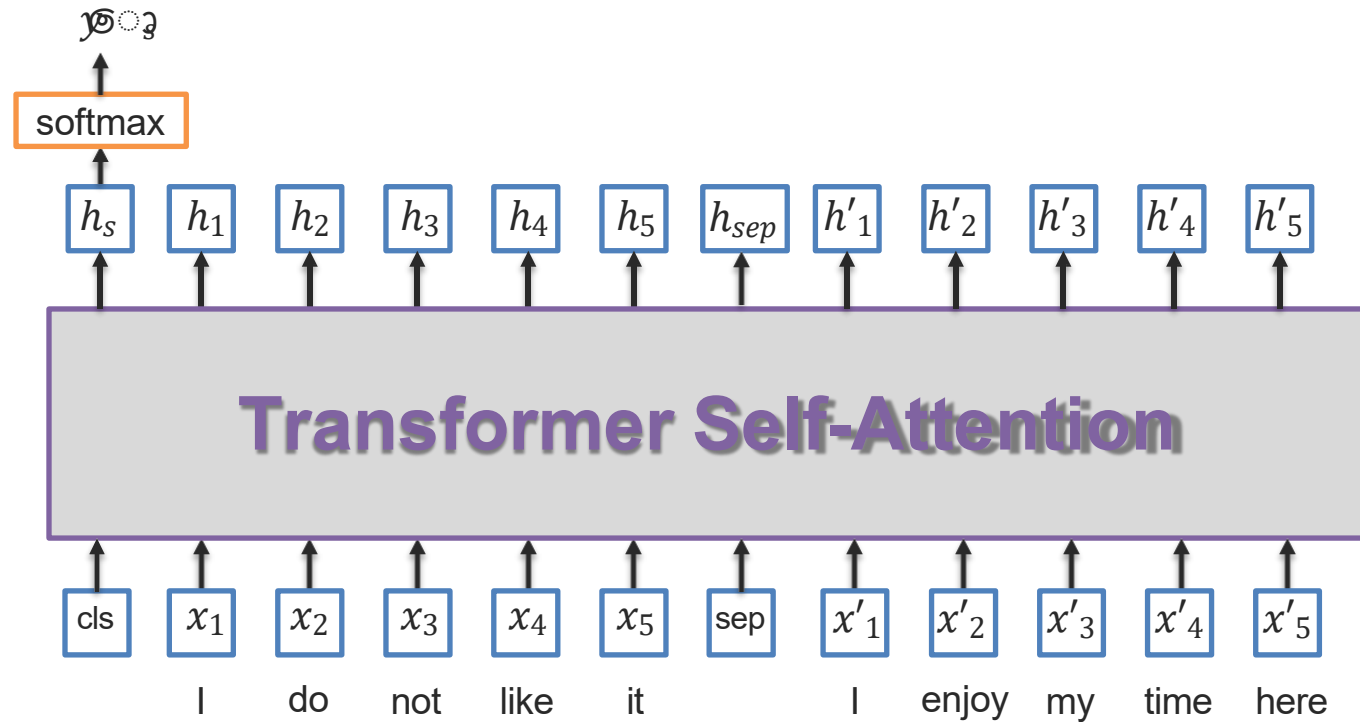


How to compare two sentences?

Fine-Tuning BERT

3 Sentence-level classification for two sentences

Examples: natural language inference



Fine-Tuning BERT

4 Question-answering: find start/end of the answer in the document

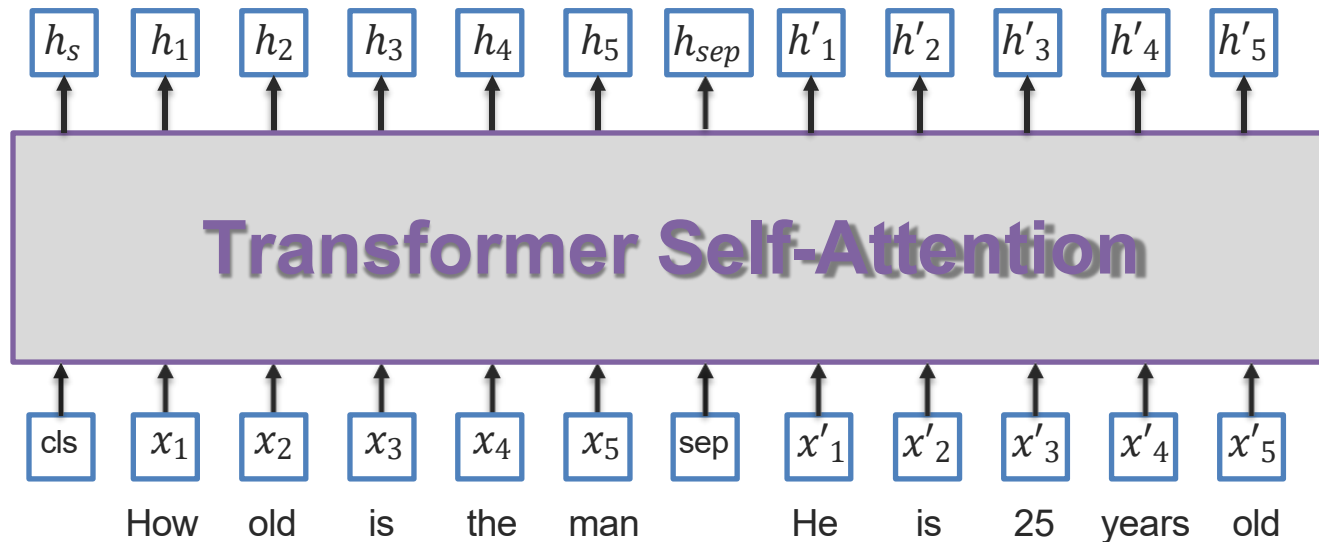
Paragraph: “... Other legislation followed, including the Migratory Bird Conservation Act of 1929, a 1937 treaty prohibiting the hunting of right and gray whales, and the Bald Eagle Protection Act of 1940. These later laws had a low cost to society—the species were relatively rare—and little opposition was raised.”

Question 1: “Which laws faced significant *opposition*?”

Plausible Answer: *later laws*

Question 2: “What was the name of the 1937 *treaty*?”

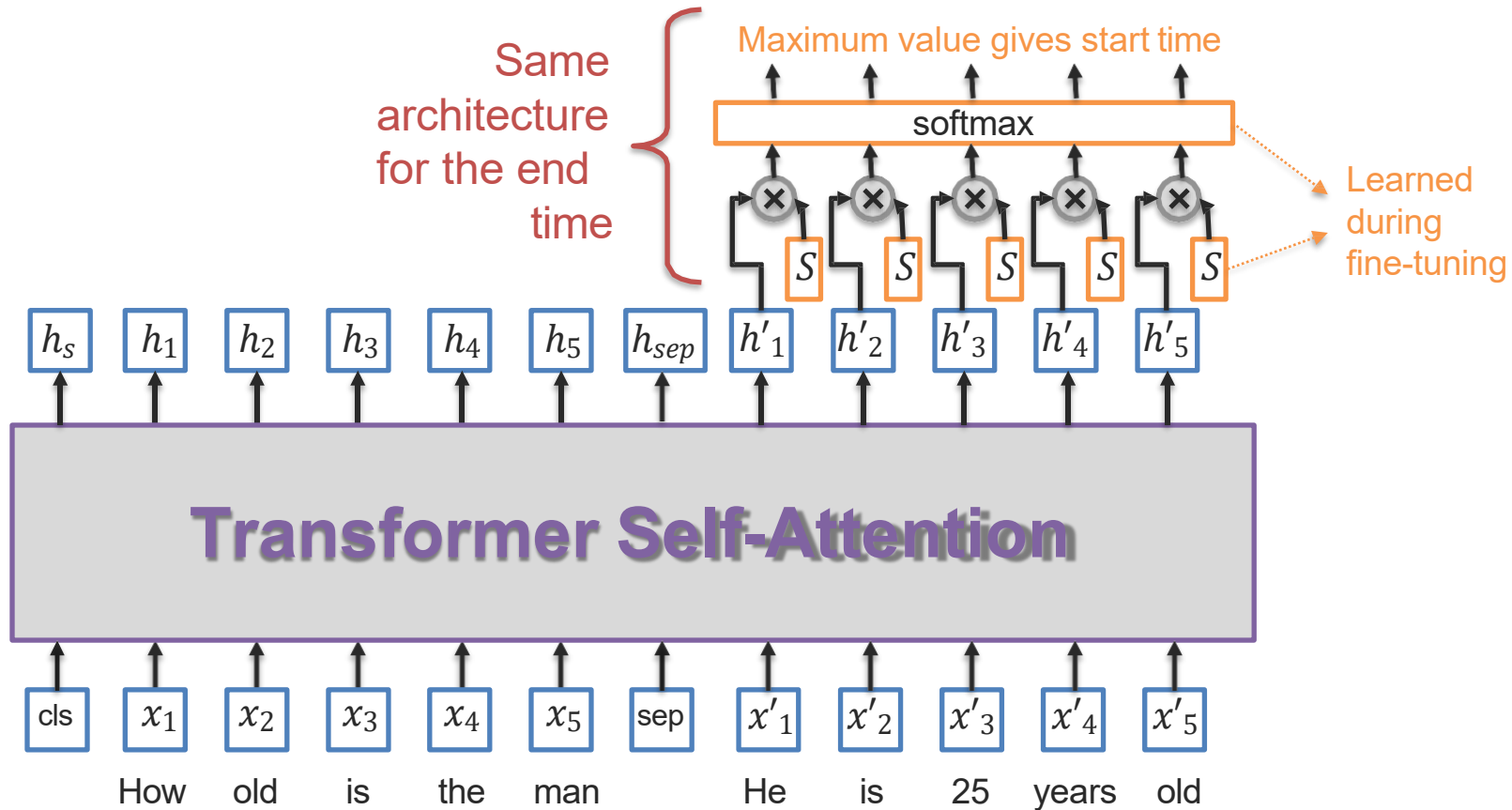
Plausible Answer: *Bald Eagle Protection Act*



How?

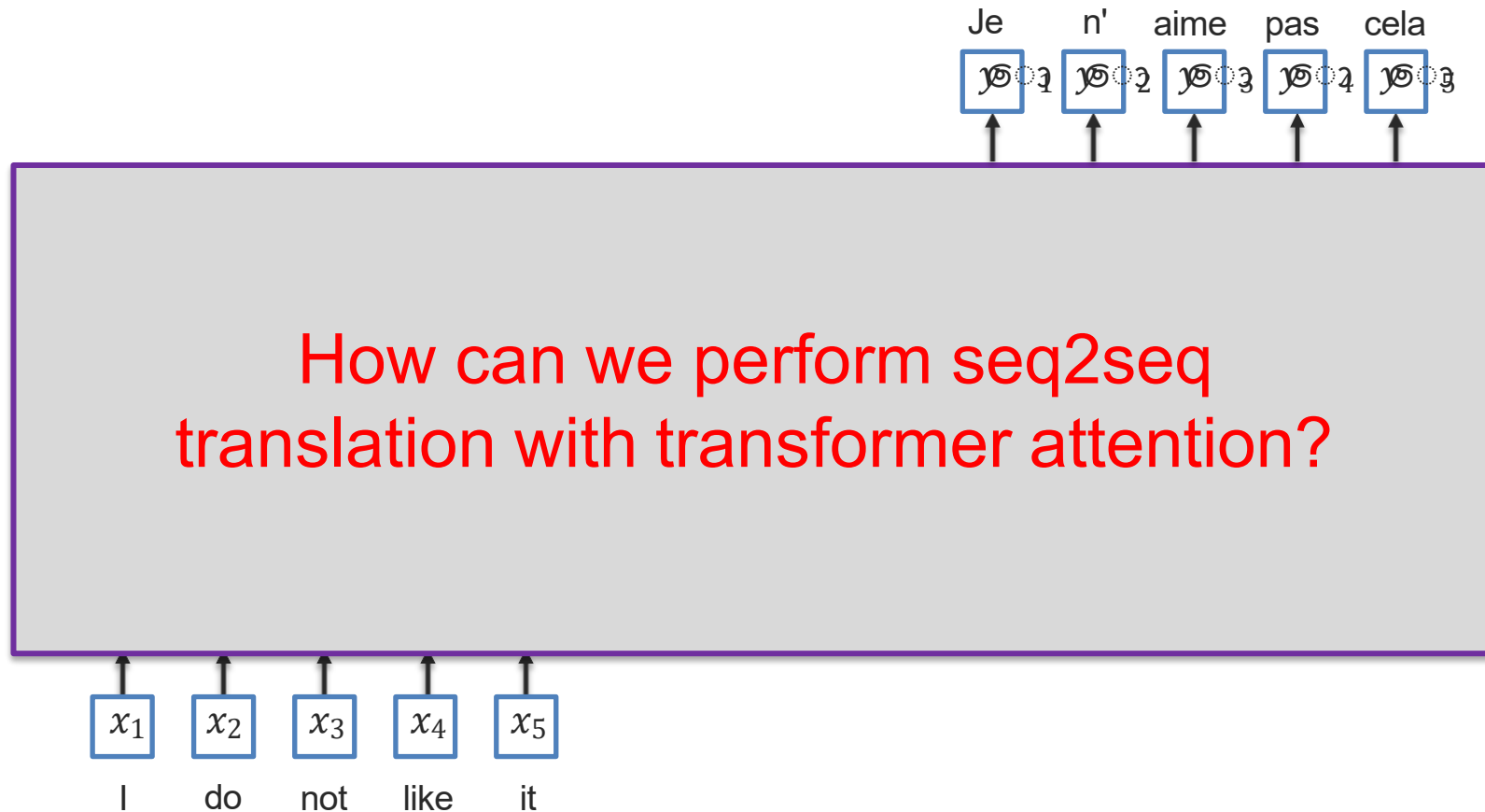
Fine-Tuning BERT

- 4 Question-answering: find start/end of the answer in the document

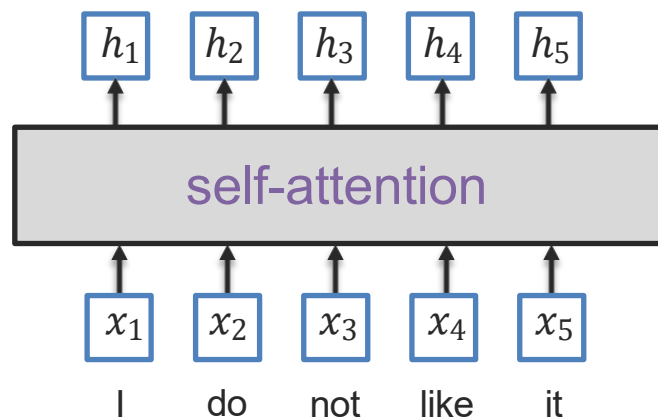
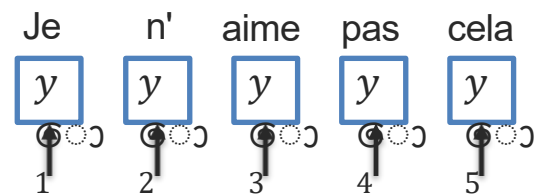


Sequence-to-Sequence Using Transformer

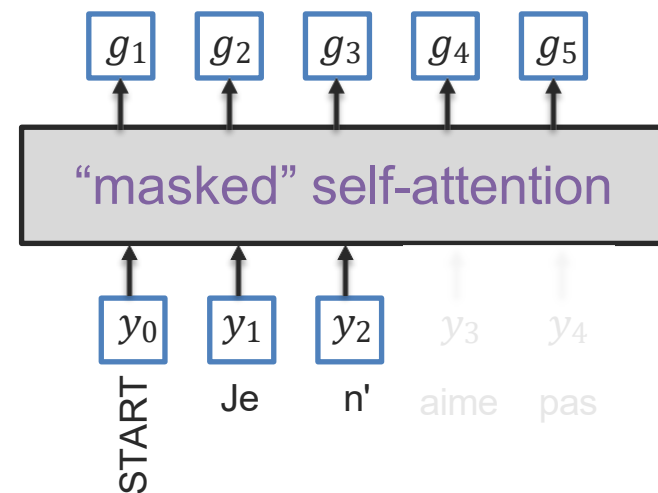
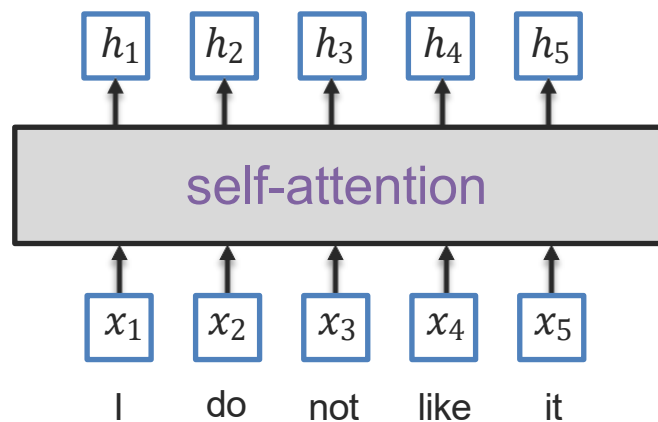
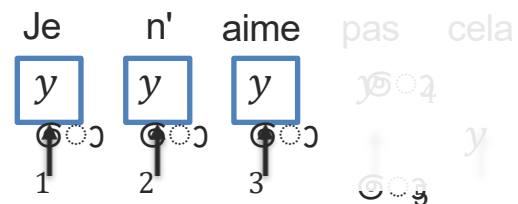
Sequence-to-Sequence Modeling



Seq2Seq with Transformer Attentions

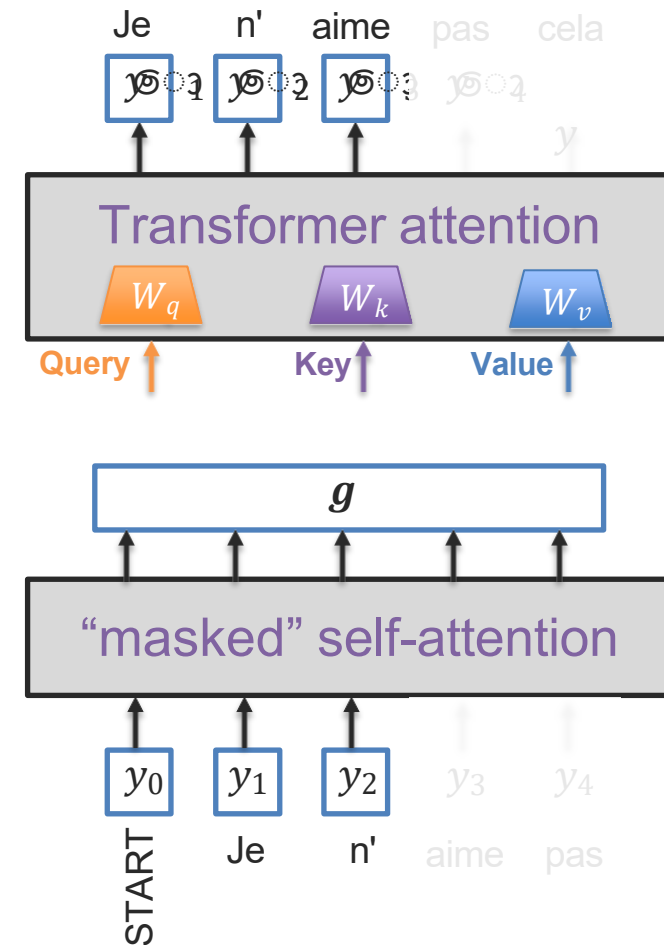
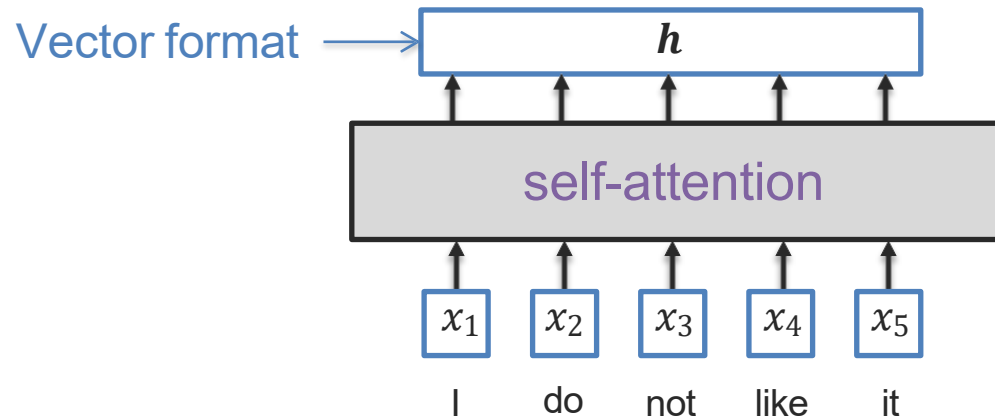


Seq2Seq with Transformer Attentions

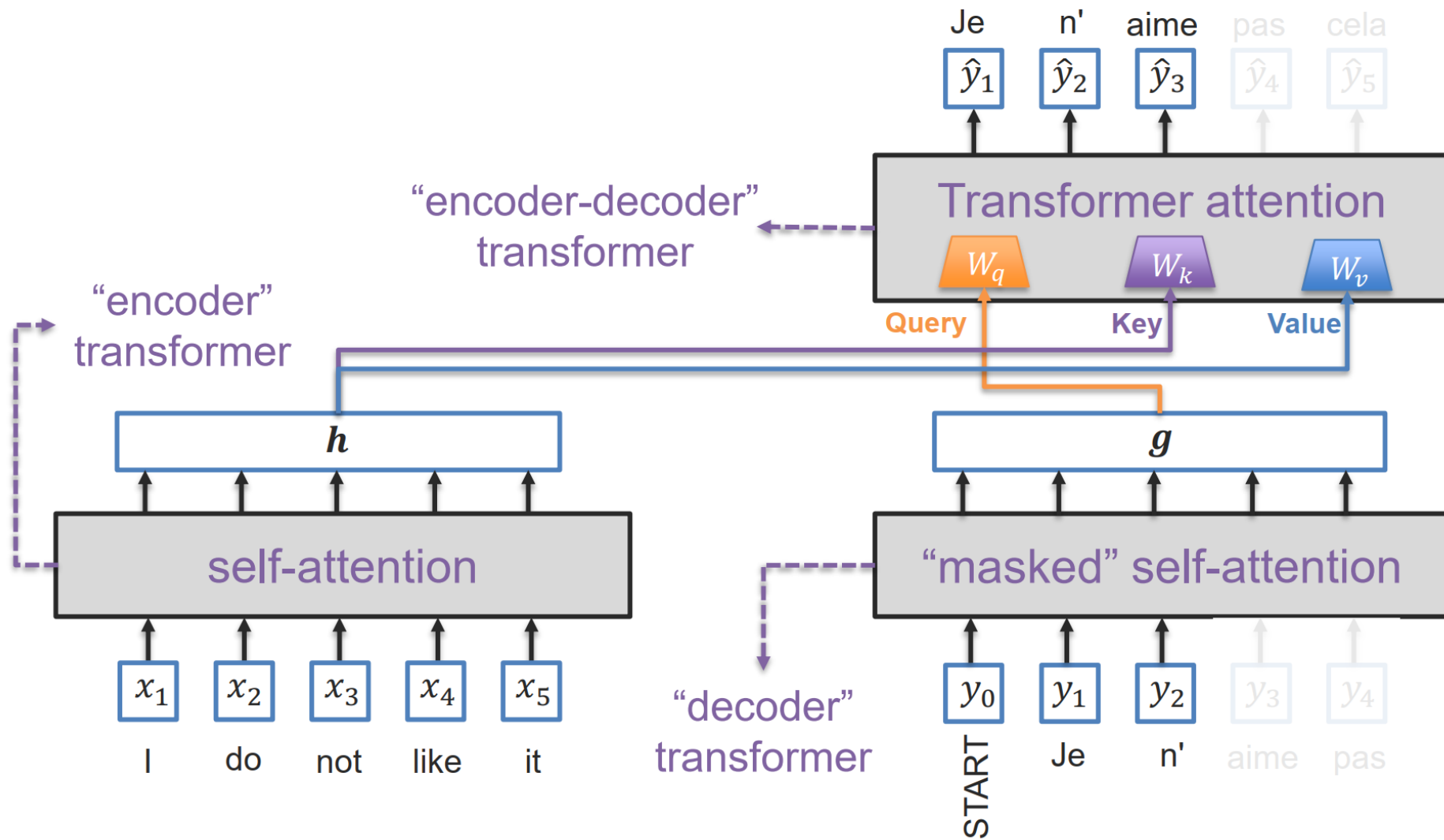


Seq2Seq with Transformer Attentions

How should we connect the encoder and decoder self-attention to the transformer attention?



Seq2Seq with Transformer Attentions



And Many More... Next 2 week!



References

- Multi Modal Machine Learning, 11-777 • Fall 2023 • Carnegie Mellon University